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Advancements in metal-dielectric multilayer structures: tailoring for enhanced optoelectronic performance

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Dielectric/metal/dielectric (D/M/D) multilayer transparent conductive structures have emerged as promising alternatives to conventional transparent conducting oxides (TCOs) for next-generation optoelectronic applications owing to their ability to simultaneously achieve high optical transparency and low electrical resistance through rational multilayer engineering. Unlike single-component transparent electrodes, D/M/D architectures exploit the synergistic interplay between ultrathin metallic interlayers and dielectric coatings to optimize the charge transport, optical interference, and interfacial stability of the electrode. Recent advances in thickness optimization, interface engineering, defect modulation, and plasmonic design have significantly expanded their applicability in photovoltaics, flexible electronics, photodetectors, transparent heaters, optical coatings, and wearable optoelectronic systems. This review presents a critical and comprehensive assessment of representative multilayer systems, including ZnO/Ag/ZnO, AZO/Ag/AZO, ITO/Ag/ITO, TiO_2/Ag/TiO_2, Nb_2O_5/Ag/Nb_2O_5, Ta_2O_5/Ag/Ta_2O_5, and emerging oxide/metal hybrid architectures. Comparative analysis reveals that Ag-based multilayers currently offer the most favourable balance between optical transmittance (>90%) and sheet resistance (<5 Ω/sq), although long-term stability, oxidation resistance, and material cost remain significant limitations of this technology. This review systematically examines fabrication approaches, including magnetron sputtering, sol–gel processing, PECVD, and ion-beam-assisted modification, emphasizing their influence on interfacial quality, scalability, and performance reproducibility. Advanced characterization methodologies, such as X-ray diffraction, X-ray photoelectron spectroscopy, atomic force microscopy, Hall measurements, Raman spectroscopy, and UV–visible analysis, are critically discussed to establish structure–property–performance correlations in the multilayer systems. Defect engineering, interfacial electronic modification, and optical bandgap tuning are effective methods for enhancing the performance of these Advanced Materials. Beyond laboratory-scale demonstrations, this review evaluates the industrial translation challenges, including the manufacturing cost, production yield, thickness uniformity, and roll-to-roll compatibility of these devices. Emerging opportunities involving flexible transparent electrodes, tandem photovoltaic architectures, plasmonic photonics, and multifunctional smart coatings are also discussed. Overall, this study provides a critical perspective on the current maturity of D/M/D multilayer technologies and identifies key research priorities, including alternative low-cost metallic interlayers, scalable deposition strategies, and interface stabilization approaches, which are

necessary for the practical commercialization of high-performance transparent optoelectronic devices.

KEYWORDS

multilayer structure, optoelectronic devices, sandwich structure, sputtering, transparent conducting oxides

1 Introduction

Transparent conducting electrodes (TCEs) are essential components of optoelectronic devices, such as solar cells, flat-panel displays, touch screens, photodetectors, and light-emitting diodes. It is difficult to find materials that have both great optical transparency in the visible range and good electrical conductivity at the same time. Transparent conducting oxides (TCOs), especially indium tin oxide (ITO), have been widely used for this purpose because they have good optoelectronic properties (Al-Kuhaili et al., 2012; Ahmad and Jaffri, 2018; Akhmedov et al., 2020; Ali et al., 2023; Chavan et al., 2023).

TCOs are a special type of wide-bandgap semiconductor (E.g., ≥ 3 eV) that allows visible light to pass through while remaining an excellent conductor. ZnO, SnO_2, and ITO are the most studied and used materials in optoelectronics because they balance optical and electrical performance (Kong and Lee, 2020; Arya et al., 2021; Barman et al., 2021). ZnO-based materials have garnered significant interest owing to their substantial exciton binding energy (~60 meV) and extensive bandgap (~3.37 eV), rendering them suitable for use in photovoltaics, sensors, and light-emitting devices (Bokov et al., 2021; Zhang et al., 2023). To achieve high conductivity in these oxide-based systems, they are often doped or the film is made thicker, which may reduce transparency and increase difficulty (Jenifer and Parthiban, 2023).

Despite its widespread use, ITO's inherent flaws of ITO have sparked the search for alternative materials. One of the most significant problems is that indium is very expensive and difficult to obtain, making it challenging to produce large quantities that are both durable and affordable. In addition, ITO films are mechanically brittle; thus, they cannot be used in flexible and wearable electronics. Furthermore, the fabrication of high-quality ITO films usually requires high processing temperatures, which increases the cost of fabrication and reduces their compatibility with substrates that are sensitive to temperature (Chavan et al., 2023). These constraints underscore the pressing necessity for alternative transparent conductive materials capable of addressing these deficiencies while preserving or enhancing their performance.

In this context, metal-dielectric multilayer structures, especially dielectric/metal/dielectric (D/M/D) topologies, have become viable options for next-generation TCEs. There are two dielectric layers, like ZnO, TiO_2, SnO_2, or Al_2O_3, on either side of an ultrathin metallic layer, which is usually Ag, Au, or Cu (Dhar and Alford, 2012; Nur-E-Alam et al., 2020; Barman et al., 2021). The metal layer has excellent electrical conductivity, whereas the dielectric layers improve optical transmittance by reducing reflection losses and protecting the metal layer from oxidation and damage from the environment. This combination enables D/M/D structures to perform better than regular single-layer TCOs in optoelectronic applications.

The thickness of each layer, choice of materials, quality of the interface, and conditions of deposition are crucial for the performance of metal-dielectric multilayer systems. The thickness of the metal layer plays an important role in determining the trade-off between the electrical conductivity and optical transparency. Ultrathin metal layers (usually less than 20 nm) can have low sheet resistance and excellent transmittance as long as the film stays continuous and homogeneous (Zhao et al., 2018; Barman et al., 2021). In addition, dielectric layers can be designed to improve light transmission by matching the refractive index and using interference effects. Recent research has shown that well-designed multilayer structures, such as ZnO/Ag/ZnO and ITO/Ag/ITO, can exhibit optical transmittance greater than 90% and sheet resistance less than 5 Ω/sq, making them strong competitors to traditional TCOs (Girtan, 2012; Wang et al., 2025).

In addition to structural optimization, new methods, including ion-beam modification, defect engineering, and plasmonic tuning, have enabled the optoelectronic properties of multilayer systems to be changed in novel ways. Ion implantation, for example, allows us to change the properties of a material in a controlled manner by adding defects and changing its electronic structure. This can make it more conductive and change how it interacts with light (Singh et al., 2018). Plasmonic effects linked to metallic layers can further improve the interaction between light and matter, making these structures even more useful (Çetinkaya et al., 2022).

Despite significant progress in creating dielectric/metal/dielectric multilayer structures, certain issues remain unresolved. Addressing issues related to large-scale production, long-term stability, interfacial diffusion, and cost-effective processing is essential for making these materials widely available for sale. Furthermore, to achieve the best performance of a device, we need to have a full understanding of how the structural parameters and optoelectronic qualities work together.

This paper focuses on recent progress in dielectric/metal/dielectric multilayer structures, specifically highlighting the dielectric/metal/dielectric configurations for optoelectronic applications. This study examines the fabrication processes, material selection strategies, and characterization methodologies used to systematically test the structural, optical, and electrical properties. This paper also provides a critical overview of the ways to improve performance, such as controlling the thickness, engineering interfaces, and modifying materials with ion beams. Finally, we discuss new uses, existing problems, and future research plans to provide a comprehensive overview of how multilayer structures can be used in next-generation optoelectronic technology. Figure 1

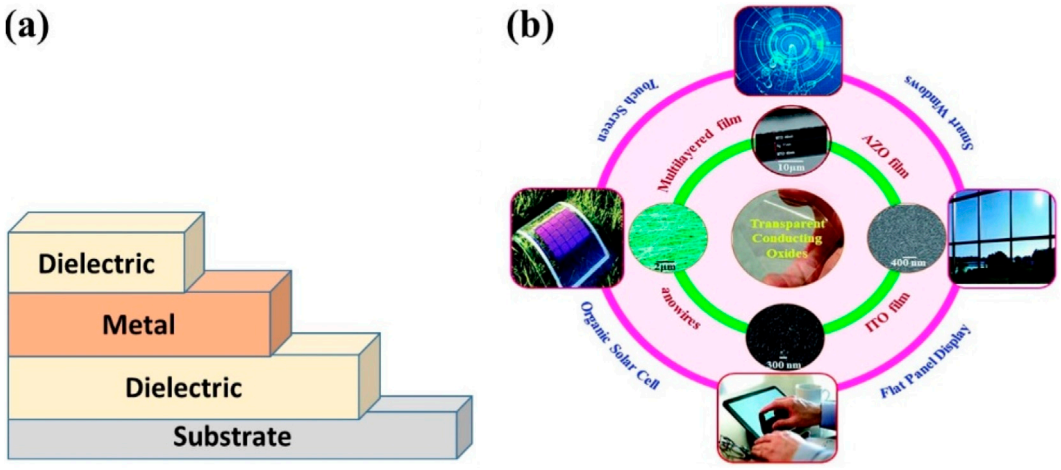


FIGURE 1
(a) Dielectric/metal/dielectric (D/M/D) multilayer structures. **(b)** Schematic illustration of the applications of different transparent conducting oxides (Dalapati et al., 2021).

shows the schematic D/M/D structure and different applications of the TCOs.

2 Dielectric/metal/dielectric multilayer structures

Dielectric/metal/dielectric multilayer structures represent a significant advancement in optoelectronic materials, enabling simultaneous optimization of optical transparency and electrical conductivity. These structures, typically configured as dielectric/metal/dielectric (D/M/D) systems, consist of an ultrathin metallic layer sandwiched between two dielectric layers. Such architectures have been widely explored as alternatives to conventional transparent conducting oxides due to their superior tunability and performance (Girtan, 2012; Sharma et al., 2016b; 2021; Sun et al., 2019).

The underlying principle of these multilayer systems is the synergistic interaction between metallic and dielectric layers. The metallic layer provides high electrical conductivity due to free electron transport, while the dielectric layers enhance optical transmission by reducing reflection losses and enabling constructive interference effects (Hong et al., 2020; Orlenson et al., 2022; Shen et al., 2022; Uyanik et al., 2022). This combination allows multilayer structures to overcome the intrinsic limitations of single-layer TCOs.

Various multilayer systems have been investigated, including ZnO/Ag/ZnO (Ho Kim et al., 2015), TiO_2/Ag/TiO_2 (Zhu et al., 2016), SnO_2/Ag/SnO_2 (Sharma et al., 2016b), ITO/Ag/ITO (Uyanik et al., 2022), and WO_3/Ag/WO_3 (Jeon et al., 2012), which have demonstrated excellent optoelectronic properties for different applications. These structures have been successfully utilized in solar cells, flat-panel displays, transparent electrodes, and heat mirrors (Al-Kuhaili et al., 2012; Sharma et al., 2016b; 2021; Li et al., 2025).

2.1 Comparative performance of multilayer structures

Table 1 quantitatively illustrates the trade-off between the optical transmittance and sheet resistance of the various metal–dielectric multilayer structures. Structures such as ITO/Ag/ITO and Ta_2O_5/Ag/Ta_2O_5 achieve an optimal balance, combining high transparency (>90%) with low sheet resistance (<5 Ω/sq) (Tang et al., 2006; Uyanik et al., 2022). In contrast, Cu-based systems, such as AZO/Cu/AZO, exhibit much higher resistance because of oxidation and lower carrier mobility (Barman et al., 2021). These results highlight the importance of material selection and thickness optimization in achieving high-performance multilayer structures.

Figure 2 illustrates the trade-off between the optical transmittance and sheet resistance in representative dielectric/metal/dielectric multilayer transparent conductive structures reported in the literature. The logarithmic scale on the Y-axis highlights the performance differences between low- and high-resistance multilayer systems.

ZnO/Ag/ZnO-based multilayers have been extensively studied because of their excellent balance between cost, performance, and environmental stability. These structures demonstrate high transmittance (~80–90%) and low sheet resistance, making them strong candidates for replacing ITO in certain applications (Barman et al., 2021; Kaya et al., 2024). Additionally, multilayers incorporating alternative dielectric materials such as Nb_2O_5, WO_3, and Al_2O_3 have shown promising results in specific applications such as anti-reflective coatings and heat mirrors (Al-Kuhaili et al., 2012; Dhar and Alford, 2012; Xiao et al., 2016).

2.2 Role of metal layer: Preference for silver (Ag)

Among the various metals used in multilayer structures, silver (Ag) is the most widely used because of its superior electrical and

TABLE 1 Trade-off between optical transmittance and sheet resistance in metal–dielectric multilayer structures.

S.No.	Multilayer structure	Optical transmittance (%)	Sheet resistance (Ω/sq)	Deposition method	Device application	References
1	ZnO/Ag/ZnO	87.0	5.4	RF magnetron sputtering	Solar cell	Barman et al. (2021)
2	AZO/Ag/AZO	80.0	4.05	RF magnetron sputtering	Solar films	Barman et al. (2021)
3	ITO/Ag/ITO	92.3	3.8	Magnetron sputtering + electro-annealing	UVLEDs	Uyanik et al. (2022)
4	Ta_2O_5/Ag/Ta_2O_5	91.64	2.53	RF sputtering	Anti-reflective coating (ARC) perovskite solar cell	Tang et al. (2006)
5	Nb_2O_5/Ag/Nb_2O_5	90.1	9.7	RF/DC sputtering	OLEDs, photodetectors	Dhar and Alford (2012)
6	WO_3/Au/WO_3	84.0	11.0	Thermal evaporation	transparent heat mirrors (THMs)	Al-Kuhaili et al. (2012)
7	TiO_2/Ag/TiO_2	85.2	6.4	Magnetron sputtering	Solar cell	Zhu et al. (2016)
8	SnO_2/Au/SnO_2	85.0	7.1	Ion beam sputtering	High-Performance Gas Sensors	Sharma et al. (2016b)
9	GZO/Ag/GZO	95.74	9.49	DC magnetron sputtering	Smart Windows and Displays	Akhmedov et al. (2020)
10	ZnO/AgOx/ZnO	89.5	5.1	Magnetron sputtering	Solar cells and photovoltaic devices	Kong and Lee (2020)
11	MoO_3/Ag/MoO_3	89.0	5.3	Thermal evaporation	Semi-Transparent Solar Cells	Sun et al. (2019)
12	ZnSnO/Au/ZnSnO	88.6	6.2	Magnetron sputtering	Displays	Jenifer and Parthiban (2023)
13	AZO/Cu/AZO	70.0	143.4	RF magnetron sputtering	Solar cell	Barman et al. (2021)

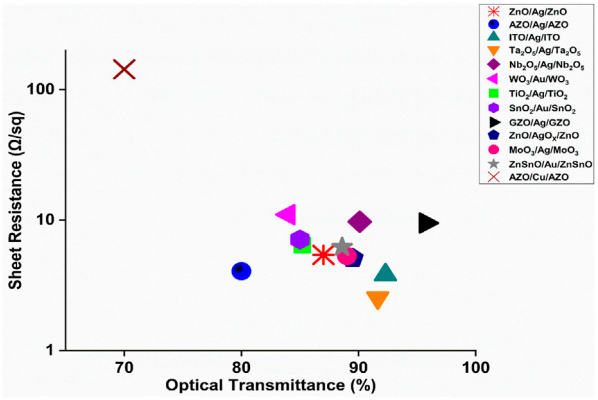


FIGURE 2
Trade-off between optical transmittance and sheet resistance.

optical properties. Ag exhibits the lowest resistivity among all metals (~1.59 μΩ cm), which enables efficient charge transport even at ultrathin thicknesses (Zhao et al., 2018). Furthermore, Ag has a relatively low optical absorption in the visible region compared to other metals, such as Au and Cu, making it highly suitable for transparent electrode applications.

Another important advantage of Ag is its ability to form continuous films at relatively low thicknesses (~10–15 nm), provided that appropriate deposition conditions are used. This is critical because discontinuities in the metal layer can lead to increased resistance and degradation of the device performance. In contrast, metals such as Cu are more prone to oxidation and may require additional protective layers, whereas Au, although chemically stable, exhibits higher optical losses and is significantly more expensive than Cu.

Therefore, Ag-based multilayer structures, such as ZnO/Ag/ZnO and ITO/Ag/ITO, have become the most

widely studied and optimized systems for optoelectronic applications to date.

2.3 Governing design principles and performance trade-offs in dielectric/metal/dielectric multilayer structures

The performance of dielectric/metal/dielectric (D/M/D) multilayer transparent conductive structures is fundamentally governed by the complex interplay between the optical transmission, electrical conductivity, interface quality, and structural continuity of the layers. Unlike conventional transparent conducting oxides, in which conductivity primarily arises from doped semiconductor carrier transport, D/M/D architectures rely on hybrid transport mechanisms that combine metallic free-electron conduction with dielectric-mediated optical interference control. Consequently, device performance cannot be understood merely through material selection; instead, it requires consideration of several coupled physical phenomena.

A central design challenge in multilayer transparent conductive structures is the intrinsic tradeoff between optical transmittance and electrical conductivity. This trade-off arises because increasing the metallic layer thickness generally improves the electrical conductivity by lowering the sheet resistance through enhanced electron transport continuity, but simultaneously increases the optical absorption and reflectance owing to free-electron interactions with incident electromagnetic radiation (Zhao et al., 2018; Barman et al., 2021).

The expanded comparative analysis presented in Figure 2 and Table 2 quantitatively demonstrates this trade-off. Representative multilayer systems such as ITO/Ag/ITO and Ta_2O_5/Ag/Ta_2O_5 exhibit excellent performance, achieving optical transmittance above 90% with sheet resistance below 5 Ω/sq, whereas systems incorporating lower-conductivity or oxidation-prone metallic layers, such as AZO/Cu/AZO, exhibit significantly higher resistance despite acceptable transparency (Tang et al., 2006; Uyanik et al., 2022). These observations emphasize that multilayer optimization requires the simultaneous balancing of optical and electrical constraints rather than maximizing either of them independently.

2.3.1 Percolation threshold and metallic film continuity

One of the most critical phenomena governing D/M/D multilayers is the metallic percolation behaviour. Since metallic interlayers are typically deposited at ultrathin thicknesses (commonly 5–15 nm), electrical transport strongly depends on whether the deposited metal forms a continuous conductive film or remains as isolated islands.

At subcritical thickness, deposited metallic atoms nucleate into discontinuous islands due to surface energy minimization rather than forming uniform continuous films. Under such conditions, electron transport is severely hindered because charge carriers must tunnel across disconnected metallic regions, resulting in a significant increase in the sheet resistance (Wu et al., 2024).

Once the metallic thickness exceeds the percolation threshold, the isolated islands coalesce into continuous conductive pathways, producing a sharp reduction in the resistivity. However, further increases in thickness introduce greater optical losses owing to enhanced free-electron absorption and reflection.

Thus, the metallic thickness must be optimized near the percolation threshold, where continuous conductivity is achieved while minimizing optical penalties. For Ag-based multilayers, this optimum typically lies in the range of 8–15 nm, depending on the substrate surface energy, deposition conditions, and interface chemistry (Zhao et al., 2018).

This principle explains why Ag remains particularly advantageous: its high intrinsic conductivity and favourable thin-film growth behaviour allow conductive continuity at a relatively low thickness compared with Cu or Au systems.

2.3.2 Plasmonic effects and optical loss mechanisms

The optical performance of D/M/D multilayer structures is also strongly influenced by plasmonic phenomena associated with the metallic interlayer. When incident electromagnetic radiation interacts with free electrons in ultrathin metallic films, collective oscillations of conduction electrons, known as surface plasmons, can be excited at the metal/dielectric interfaces (Thallapally et al., 2008). These plasmonic interactions may be beneficial in certain photonic or sensing applications by enhancing local electromagnetic fields, increasing light–matter interactions, and enabling tailored spectral responses. However, in transparent conductive electrode applications, plasmonic behaviour can introduce parasitic optical losses through the following mechanisms:

- increased free-carrier absorption,
- enhanced reflection,
- non-radiative energy dissipation.

As the metallic thickness increased, the plasmonic absorption became stronger, reducing the visible transmittance. Therefore, optical optimization requires balancing the plasmonic enhancement effects with the transparency loss. This trade-off is particularly important for multilayers intended for photovoltaic or display applications, where excessive plasmonic absorption reduces device efficiency.

2.3.3 Interface scattering and carrier transport

In addition to thickness optimization, the interface quality is another dominant factor that determines the performance of multilayers. Charge carriers moving through D/M/D architectures encounter multiple dielectric/metal interfaces, each of which may introduce scattering, trapping, or recombination depending on the structural quality. Rough surfaces make it more likely for carriers to scatter, which lowers their mobility and increases energy losses.

Similarly, atomic interdiffusion between metallic and dielectric layers can modify the interface chemistry, alter the band alignment, and generate defect states that degrade the charge transport.

Interface-related performance degradation may arise from the following factors:

TABLE 2 Comparative analysis of fabrication techniques.

Methods	Advantages	Disadvantages
1. Sputtering	1. High quality film 2. Ultra-thin layer 3. Low-temperature processing 4. Scalability 5. *In situ* doping using stoichiometry	1. High cost 2. Surface damaged by high-power deposition 3. Composition
2. Atomic layer deposition (ALD)	1. Low-temperature processing 2. Excellent adhesion 3. High-quality film 4. Ultra-thin films	1. Large area deposition 2. Economic viability 3. Limitation of materials 4. Time processing
3. Chemical vapor deposition	1. Ultra-thin film 2. Epitaxial growth 3. High crystal-quality 4. Stoichiometry	1. costly and difficult procedure 2. Scalability
4. Thermal evaporation	1. Cost-effectiveness and reliable film quality 2. Synthesis of uniformly thick coatings across a large area	1. High-temperature processing 2. Low environmental stability 3. Mechanical durability
5. Spin coating	1. Quick deposition 2. able to create a suitable thickness film 3. Easy integration 4. Easy synthesis	1. Scalability 2. Dopant incorporation 3. Lack of material efficiency 4. Material wastage
6. Hydro/solvothermal	1. Large-scale synthesis 2. Easy technique 3. Low cost	1. Safety issues 2. Ultra-thin film 3. Long-time reaction 4. Crystal quality
7. Sol-gel process	1. Reproducibility 2. Low temperature sintering process 3. Low cost 4. Simple technique 5. Achieve proper stoichiometry	1. Large-scale production 2. Interface and bulk defect existence 3. Synthesis of ultra-thin films (<10 nm)
8. Spray coating	1. Excellent consistency in the composition 2. Easy, profitable, and affordable 3. Large-area coating	1. Energy consumption 2. Synthesis of ultra-thin film 3. Low conductivity 4. Wastage of solution
9. Chemical bath deposition	1. Easy synthesis 2. Low-temperature process 3. Grain size and film structure tenability 4. Multiple/single precursor 5. Large-area production 6. Strong adhesive	1. Bulk and interface defects are present 2. Long period of deposition 3. Various doping designs and synthesis 4. Synthesis of ultra-thin films (<10 nm) 5. Requirement of seed layer
10. Inkjet printing	1. Digital control and design flexibility 2. Precise material deposition 3. Waste reduction	1. Ink formulation 2. Limited material selection 3. Requirement for heat treatment and post processing

- roughness-induced scattering,
- contamination during deposition,
- interfacial oxidation,
- atomic intermixing,
- vacancy accumulation.

These effects become increasingly important as the layer thickness decreases because the interface contributions become comparable to those of bulk transport. Therefore, high-performance multilayer designs require atomically smooth interfaces, chemically stable layer boundaries, and minimal contamination levels.

2.3.4 Optical interference engineering

Unlike conventional single-layer conductive films, D/M/D multilayers offer unique optical tunability via interference engineering. Dielectric layers function not only as protective coatings but also as optical interference modulators. By carefully selecting the dielectric thickness and refractive index, constructive interference in the visible spectrum can be promoted, while reflection losses are minimized (Sharma et al., 2021).

This optical design principle allows the multilayers to reduce the natural optical absorption caused by the metallic interlayer.

The effective selection of a dielectric layer requires consideration of the following factors:

- refractive index matching,
- extinction coefficient,
- wavelength-dependent interference behaviour,
- thermal and chemical stability.

High-index dielectric materials, such as TiO_2 and Ta_2O_5, often improve anti-reflective performance, whereas ZnO-based systems offer balanced optical and electronic functionalities. Thus, the dielectric selection is not merely structural but fundamentally optical in nature.

2.3.5 Practical design guidelines for high-performance multilayers

Based on the comparative analysis and governing physics, several practical design guidelines are proposed.

- Metallic layer selection

Preferred: Ag (best conductivity/transparency balance).

Alternatives: Au (stable but costly) and Cu (economical but oxidation-prone).

- Metallic thickness

Optimal: ~8–15 nm.

Too thin: discontinuous film, tunnelling transport, high resistance.

Too thick: optical absorption, reflectance increase.

- Dielectric layer design

Preferred characteristics: high transparency, appropriate refractive index, thermal stability, and chemical compatibility.

Examples: ZnO, TiO_2, Ta_2O_5, Nb_2O_5

- Interface engineering

Required: low roughness, minimal contamination, suppressed interdiffusion, and oxidation control.

- Deposition strategy

Preferred: vacuum-based precision deposition.

Examples: magnetron sputtering, ALD, and PECVD.

2.3.6 Critical perspective

Despite substantial progress, many reported multilayer studies remain system-specific optimizations rather than generalized, design-driven investigations of the underlying mechanisms. A major challenge in this field is the absence of universally transferable design rules due to the sensitivity to deposition conditions, interface chemistry, and substrate-dependent growth behaviour. Therefore, future research should increasingly focus on predictive design frameworks that integrate the following:

- experimental optimization,
- computational optical modelling,
- interface thermodynamics,
- machine-learning-assisted multilayer engineering.

Such approaches will accelerate the rational development of next-generation transparent conductive multilayer architectures.

3 Fabrication techniques for dielectric/metal/dielectric (D/M/D) multilayer structures

The fabrication methodology of dielectric/metal/dielectric (D/M/D) multilayer structures is not merely a processing route but a critical determinant of the resulting optoelectronic performance of devices. The continuity of ultrathin metallic interlayers, interfacial smoothness, defect density, adhesion strength, and thickness uniformity are strongly governed by the deposition conditions, which directly influence optical transparency, electrical conductivity, and long-term device reliability. Therefore, fabrication techniques should be evaluated not only based on their operational principles but also in terms of their influence on the structure–property modelling relationships of multilayer transparent conductive systems. For instance, precise thickness control is particularly important because metallic interlayers, such as Ag, typically operate near the electrical percolation threshold, where even sub-nanometer deviations can drastically alter the conductive continuity and sheet resistance (Kong and Lee, 2020; Uyanik et al., 2022).

3.1 RF/DC magnetron sputtering

Magnetron sputtering remains one of the most effective fabrication techniques for high-performance D/M/D multilayer transparent conductive structures due to its exceptional control over ultrathin film thickness, interface quality, and deposition uniformity. This level of precision is essential because the electrical performance of ultra-thin metallic interlayers strongly depends on the formation of continuous conductive pathways. If the metallic thickness remains below the percolation threshold, isolated island formation may dominate, resulting in a significant increase in the sheet resistance. Controlled sputtering allows the optimization of metallic continuity while minimizing excessive optical absorption. Furthermore, sputtered films generally exhibit improved interface smoothness,

reduced defect density, and enhanced adhesion, all of which contribute to lower carrier scattering and improved optoelectronic performance (Dhar and Alford, 2012; Uyanik et al., 2022). However, despite these performance advantages, sputtering remains capital-intensive because of the vacuum infrastructure, target replacement requirements, and high energy consumption.

3.2 Spin coating technique and sol–gel process

Spin coating is a simple and cost-effective technique that is widely used to deposit uniform dielectric thin films, particularly in laboratory-scale research. In this method, a precursor solution is dispensed onto a substrate, which is then rotated at high speed, allowing the solution to spread uniformly across the surface due to the centrifugal force. This technique offers advantages such as simplicity, low cost, and ease of integration with other processes. It is particularly suitable for depositing oxide-based dielectric layers such as ZnO and TiO_2. For example, multilayer structures such as $SrTiO_3/BaTiO_3$ have been successfully fabricated using spin-coating techniques (Pontes et al., 2001).

The sol–gel process is a versatile chemical technique used to synthesize high-quality dielectric thin films. This process involves the hydrolysis and condensation of metal alkoxide precursors to form a colloidal solution (sol), which subsequently evolves into a gel that is then dried and annealed to form a thin film (Bokov et al., 2021). This method allows for precise control over the composition, stoichiometry, and microstructure. It is particularly advantageous for producing uniform films at relatively low processing temperatures and is cost-effective compared to vacuum-based techniques.

Although solution-based fabrication methods, such as sol–gel processing and spin coating, offer advantages, including lower capital costs, process simplicity, and potential scalability, their applicability to high-performance multilayer transparent conductive systems remains limited by inferior thickness precision and reduced interface reproducibility. In multilayer architectures requiring nanometer scale control of metallic continuity, a non-uniform thickness distribution can lead to discontinuous conduction pathways, optical scattering losses, and poor reproducibility. Solvent-induced defects and imperfect dielectric/metal interfaces may further degrade charge transport behaviour. Consequently, despite their economic attractiveness, solution-based methods are often less competitive than vacuum-based deposition routes for high-performance transparent conductive multilayer fabrication (Chavan et al., 2023; Dong et al., 2025).

3.3 Plasma-enhanced chemical vapor deposition (PECVD)

PECVD is a widely used technique for depositing dielectric thin films, particularly in microelectronic and optoelectronic applications. In this process, plasma is used to enhance the chemical reactions of precursor gases, enabling film deposition at relatively low substrate temperatures (Nagai et al., 2003).

One of the major advantages of PECVD is its ability to produce high-quality, dense films with excellent uniformity and step coverage. This method is particularly suitable for depositing insulating layers with controlled dielectric properties.

However, PECVD systems are relatively complex and costly, and the process may involve hazardous precursor gases such as silane. Additionally, controlling the film composition and minimizing defects require careful optimization of the process parameters.

3.4 Low-energy ion beam modification

Low-energy ion beam techniques are increasingly being used as post-deposition modification tools to tailor the properties of multilayer structures for various applications. In this approach, energetic ions are implanted into the material, causing atomic displacements and defect formation, which can significantly alter the structural, optical, and electrical properties.

Ion implantation offers precise control over the dopant concentration, depth, and distribution, making it a powerful tool for engineering material properties at the nanoscale. Studies have shown that ion irradiation can induce defect states, modify the bandgap, and improve the electrical conductivity in multilayer structures such as $TiO_2/Ag/TiO_2$ (Singh et al., 2018).

Moreover, ion beam modification can improve the interface quality and enhance film densification. However, excessive irradiation may lead to structural damage, amorphization, or degradation of optical properties, necessitating the careful optimization of ion energy and fluence.

3.5 Comparative analysis of fabrication techniques

A comparative evaluation of the aforementioned fabrication techniques revealed that each method offers unique advantages depending on the intended application and required film properties.

- Magnetron sputtering remains the most preferred technique for multilayer structures due to its ability to produce high-quality, uniform, and ultrathin films with excellent reproducibility and scalability.
- Sol–gel and spin coating are attractive for low-cost and solution-based processing; however, they are limited in achieving high-performance multilayers with precise thickness control.
- PECVD is highly effective for depositing dielectric layers with controlled properties; however, it involves higher complexity and cost.
- Ion beam techniques serve as powerful tools for post-deposition modification, enabling the fine-tuning of material properties beyond the limits of conventional fabrication.

The selection of a fabrication technique plays a decisive role in determining the performance of metal–dielectric multilayer structures. Although solution-based methods offer cost advantages, vacuum-based techniques, such as sputtering, provide superior film quality and interface control, which are critical for achieving high-performance optoelectronic devices. Figure 3 shows schematic diagrams of the different deposition techniques.

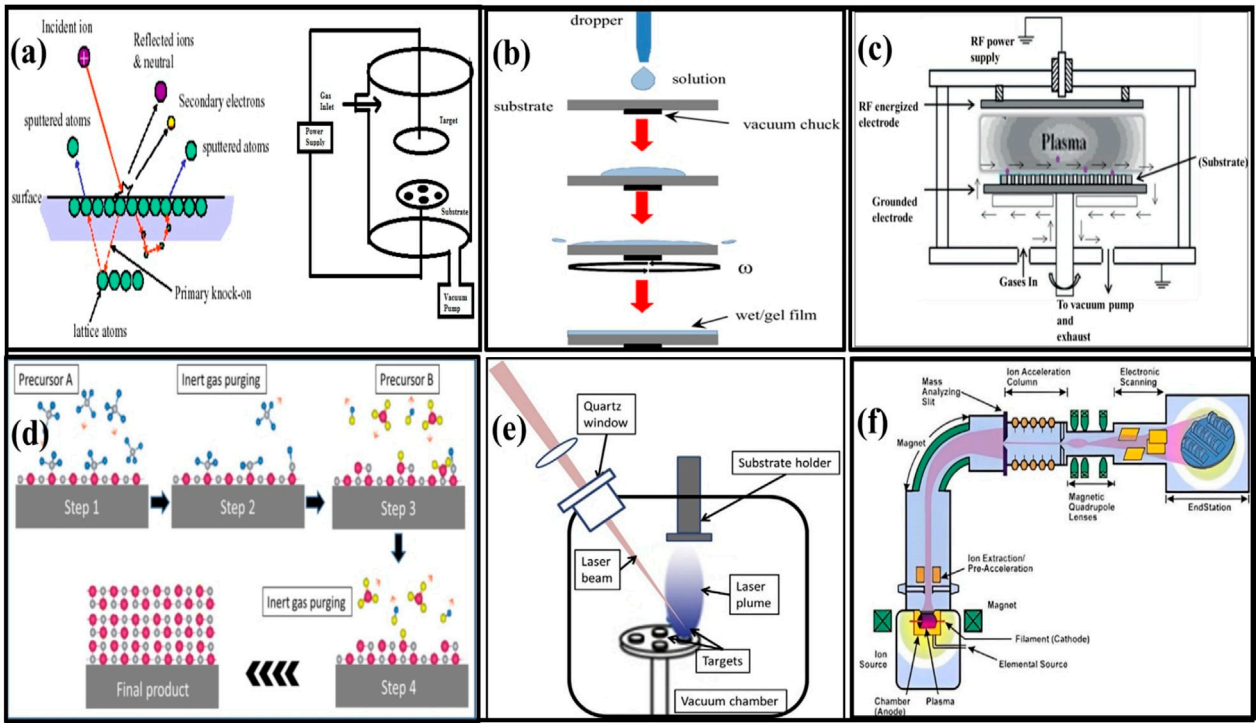


FIGURE 3
Schematic diagrams of the working of **(a)** sputtering, **(b)** spin coating, and **(c)** PECVD (Plasma Enhanced Chemical Vapor Deposition). **(d)** ALD (Atomic Layer Deposition) **(e)** pulsed laser deposition (PLD) **(f)** ion implantation.

3.6 Large-scale manufacturing challenges

Despite remarkable progress in dielectric/metal/dielectric (D/M/D) multilayer transparent conductive structures, their transition from laboratory-scale demonstrations to industrial-scale manufacturing remains challenging. Although many deposition techniques, particularly vacuum-based methods such as magnetron sputtering, have demonstrated excellent control over film thickness, interface quality, and optoelectronic performance, large-scale commercialization requires overcoming significant constraints related to fabrication cost, production yield, and thickness uniformity across extended substrate areas (Barman et al., 2021; Wang et al., 2021; Triolo et al., 2025).

3.6.1 Cost constraints and economic feasibility

One of the major barriers to the large-scale manufacturing of D/M/D multilayer structures is the high capital and operational costs associated with advanced deposition technologies. Magnetron sputtering, which is widely regarded as the most effective technique for producing high-quality multilayer transparent electrodes, requires sophisticated vacuum infrastructure, high-purity targets, controlled gas environments, and continuous process monitoring, all of which contribute substantially to manufacturing cost (Chen et al., 2019; Bandyopadhyay et al., 2020).

The use of noble metals, such as silver (Ag) and gold (Au), further increases production expenses. Silver remains the most preferred metallic interlayer because of its exceptionally low electrical resistivity and favorable optical transparency; however, its high market cost and limited resource availability raise concerns regarding its long-term economic sustainability (Zhao et al., 2018). Although gold-based multilayer systems are chemically stable, they are even less attractive for large-scale deployment because of their significantly higher material costs (Al-Kuhaili et al., 2012). In addition, sputtering targets require periodic replacement, and target poisoning in reactive deposition processes can reduce operational efficiency and increase maintenance requirements (Bandyopadhyay et al., 2020).

Energy consumption is also an important economic concern. Vacuum generation, plasma maintenance, substrate heating, and post-deposition annealing contribute to elevated energy demand, particularly when scaling production to industrial throughput levels. Consequently, although sputtering-based D/M/D fabrication delivers excellent device performance, its economic viability for low-cost commercial applications remains under active investigation (Wang et al., 2021).

3.6.2 Yield and process reliability

The production yield is another critical challenge in the manufacturing of multilayer transparent electrodes. Since D/M/D architectures rely on ultrathin metallic layers, typically in the thickness range of 5–15 nm, process deviations can significantly affect film continuity and electrical performance (Li et al., 2025).

One of the most common yield-limiting issues is the incomplete continuity of the metal film. During the deposition

of ultrathin Ag or Au layers, island-like nucleation behavior may occur instead of continuous film formation, resulting in disconnected conductive pathways and a dramatic increase in sheet resistance (Nguyen et al., 2022). These discontinuities can severely compromise the reproducibility of the device.

Other yield-related defects include.

- pinhole formation,
- thickness fluctuations,
- interfacial contamination,
- residual stress-induced cracking,
- delamination during thermal cycling.

Interface contamination is particularly problematic because D/M/D structures strongly rely on precise interfacial engineering for optimal charge transport and optical interference behavior (Yuan et al., 2022). Even trace contamination between the dielectric and metallic layers may degrade adhesion, increase scattering losses, and reduce carrier mobility.

Batch-to-batch reproducibility becomes increasingly difficult during industrial scaling because of subtle variations in deposition power, chamber pressure, target erosion, and substrate positioning. These factors collectively reduce the manufacturing yield and increase the production cost (Sharmile et al., 2025).

3.6.3 Thickness uniformity and large-area scalability

Achieving uniform multilayer deposition across large substrate areas remains one of the most technically demanding challenges in commercialization. Laboratory-scale samples often demonstrate excellent optoelectronic performance under highly controlled deposition conditions; however, translating these results to large-area substrates introduces substantial complexity (Zhao et al., 2026).

The performance of D/M/D multilayers is highly sensitive to thickness variations, especially for metallic interlayers, where even sub-nanometer deviations can significantly alter the optical transmittance, plasmonic response, and sheet resistance (Zeng et al., 2024). Maintaining such stringent thickness control across meter-scale substrates is difficult because of the following reasons:

- non-uniform plasma density,
- target erosion effects,
- substrate heating gradients,
- deposition angle variations,
- chamber geometry limitations.

Edge effects are another important consideration, as films deposited near the substrate boundaries often exhibit thickness deviations relative to the central region, resulting in non-uniform electrical and optical behavior.

These challenges become even more pronounced in flexible electronics and roll-to-roll manufacturing. Flexible polymer substrates, such as PET and PEN, are mechanically sensitive and often exhibit thermal deformation, limiting their compatibility with conventional high-temperature vacuum deposition techniques (Yakimets et al., 2010; Skafi and Brown, 2025). Although roll-to-roll sputtering offers a promising route for industrial production, achieving nanometer-scale thickness precision and interfacial consistency under continuous high-speed manufacturing remains challenging (Arun et al., 2025).

Recent efforts have explored alternative low-cost solution-based methods and hybrid deposition approaches to improve scalability; however, these methods often struggle to match the structural precision and performance reproducibility achieved by vacuum-based techniques (Musselman et al., 2016; Parida et al., 2026).

3.6.4 Outlook for industrial translation

To accelerate the commercialization of D/M/D multilayer structures, future research should focus on scalable low-cost manufacturing strategies, reduced dependence on noble metals, improved nucleation control for ultrathin continuous metallic films, and roll-to-roll compatible deposition methodologies. The development of alternative conductive materials, such as Cu-protected architectures, oxide-metal hybrid systems, and advanced interface engineering approaches, may provide practical pathways toward economically viable large-area production.

Overall, while dielectric/metal/dielectric multilayer structures demonstrate outstanding laboratory-scale performance, overcoming the challenges associated with cost, yield, and uniformity remains essential for their successful industrial deployment (Pastuszak and Węgierek, 2022).

Overall, the fabrication strategy fundamentally determines the multilayer performance through its influence on metallic film continuity, interface chemistry, structural defects, and thickness precision. Because conductivity, transparency, and durability emerge from these interconnected structural factors, the deposition methodology should be regarded as an active performance-engineering parameter rather than merely a fabrication choice (Kong and Lee, 2020; Jenifer and Parthiban, 2023).

4 Characterization methods for assessing performance

The characterization of D/M/D multilayer structures extends beyond conventional structural or compositional verification and serves as a critical framework for understanding the structure–property–performance relationships. Since multilayer optoelectronic behavior depends strongly on interface chemistry, defect evolution, structural quality, and optical loss mechanisms, characterization techniques provide direct insight into the physical origins of conductivity, transparency, and device stability. Accordingly, these methods should be interpreted as performance diagnostic tools rather than merely analytical characterization approaches (Jenifer and Parthiban, 2023).

4.1 X-ray diffraction (XRD): structural evolution and crystallinity

X-ray diffraction (XRD) provides important information regarding the crystallinity, phase evolution, lattice strain, and structural disorder in multilayer systems. These structural

characteristics directly influence the charge transport behaviour, as improved crystallinity may enhance carrier mobility by reducing grain-boundary scattering, whereas structural disorder and strain may introduce defect states that adversely affect conductivity. Thus, XRD analysis contributes directly to the understanding of the multilayer electrical performance rather than simply confirming phase formation (Dhar and Alford, 2012).

Instead of merely identifying phases, XRD plays a key role in understanding how the metal layer thickness influences the structural properties. For example, in ZnO/Ag/ZnO multilayers, the presence of characteristic ZnO (002) and Ag (111) peaks confirms the coexistence of hexagonal and face-centered cubic structures, respectively (Singh et al., 2018). Variations in the peak intensity and full width at half maximum (FWHM) with increasing Ag thickness indicate changes in the crystallite size and lattice strain.

An important observation in multilayer systems is that increasing the metal thickness can lead to enhanced peak intensity but may also introduce lattice distortion or defect formation, particularly at interfaces. This directly affects the carrier transport and optical properties. Thus, XRD analysis is a critical tool for optimizing the layer thickness and improving the structural quality.

Bragg's law, which describes the diffraction condition, is as follows:

$$2d\sin\theta = n\lambda \quad (1)$$

where d is the interplanar spacing, θ is the diffraction angle, n is the order of diffraction, and λ is the wavelength of the X-ray.

The d values were determined using the above Equation 1 with known values of θ, n, and λ. To determine the unknown material, X-ray diffraction data was obtained, printed in a table format on paper, and compared with data from the Joint Committee Power Diffraction Standards (JCPDS). Powder, single-crystal, or thin-film samples may be employed.

Using Scherrer's formula in Equation 2, the full width at half maximum (FWHM) of the most intense diffraction line is used to determine the crystallite size of the deposits as

$$D = \frac{K\lambda}{\beta \cos\theta} \quad (2)$$

Where D is crystallite size, K is Scherrer constant, λ is wavelength of X-ray, β is full width at half maxima of the peak (FWHM) in radians, and θ is Bragg's angle.

Figures 4a–d shows the XRD patterns of ZnO (200 nm)/Ag/ZnO (50 nm) multilayer thin films deposited with Ag thicknesses of 5, 10, 30, and 40 nm. These plots show that the diffraction peaks corresponding to ZnO (002) and Ag (111) become progressively sharper with increasing Ag thickness, indicating improved crystallinity and enhanced metallic continuity. The increase in Ag peak intensity suggests better crystal growth, which contributes to lower electrical resistance while maintaining good optical transparency (Thabit et al., 2021).

4.2 X-ray photoelectron spectroscopy (XPS): Interface chemistry and defect states

X-ray photoelectron spectroscopy (XPS) is particularly important for multilayer systems because the dielectric/metal interfaces strongly govern the electronic transport, chemical stability, and defect-mediated performance. XPS enables the analysis of oxidation states, interfacial diffusion, chemical bonding, and defect-state formation. For example, the oxidation of ultrathin Ag interlayers may significantly increase the electrical resistance, whereas oxygen vacancy formation in oxide layers may alter the carrier concentration and optical absorption behaviour. Therefore, interfacial chemical instability may directly compromise both conductivity and long-term device reliability (Kong and Lee, 2020; Jenifer and Parthiban, 2023).

Furthermore, XPS analysis can confirm the metallic nature of the embedded metal layer (e.g., Ag^0) and detect any undesirable oxidation (Ag_2O or AgO), which can degrade the performance. Therefore, XPS is essential for ensuring chemical stability and optimizing the interface quality of multilayer structures. The XPS findings are illustrated in Figures 5A–D.

4.3 Atomic force microscopy (AFM): Surface morphology and roughness control

Atomic force microscopy (AFM) provides critical information regarding nanoscale surface morphology and roughness, both of which strongly affect the performance of multilayers. Increased roughness may enhance electron scattering, reduce carrier mobility, weaken interfacial adhesion, and increase optical scattering loss. In ultrathin multilayer architectures, where interfacial smoothness strongly influences conductive continuity and optical transparency, AFM serves as an important performance diagnostic tool (Uyanik et al., 2022).

Multilayer systems require smooth and continuous surfaces to ensure efficient current flow and to minimize scattering losses. AFM studies have shown that ion irradiation can lead to surface smoothing and reduced roughness, thereby improving electrical performance (Crespo-Sosa et al., 2003). For example, a reduction in RMS roughness from ~2.97 nm to ~2.48 nm has been observed with increasing ion fluence, indicating improved film uniformity.

Importantly, AFM analysis helps identify issues such as island formation or discontinuities in ultra-thin metal layers, which can significantly increase the sheet resistance. Thus, AFM is a key tool for optimizing the deposition conditions and ensuring film continuity. Figures 6A–D shows the AFM images of the pristine and SHI-irradiated ZAZ multilayers. The pristine film exhibited relatively larger surface grains and higher surface roughness. After ion irradiation, the grain distribution became more homogeneous, and the RMS roughness decreased progressively, indicating surface smoothing caused by irradiation-induced atomic rearrangement. Such smooth interfaces reduce carrier scattering and improve electrical transport (Sharma et al., 2017).

4.4 UV–visible spectroscopy: optical behaviour and bandgap engineering

UV–visible spectroscopy directly measures the transparency of a material, the amount of light it absorbs, and the behaviour of its optical bandgap in multilayer systems. In addition to transmittance analysis, UV–visible characterization helps identify

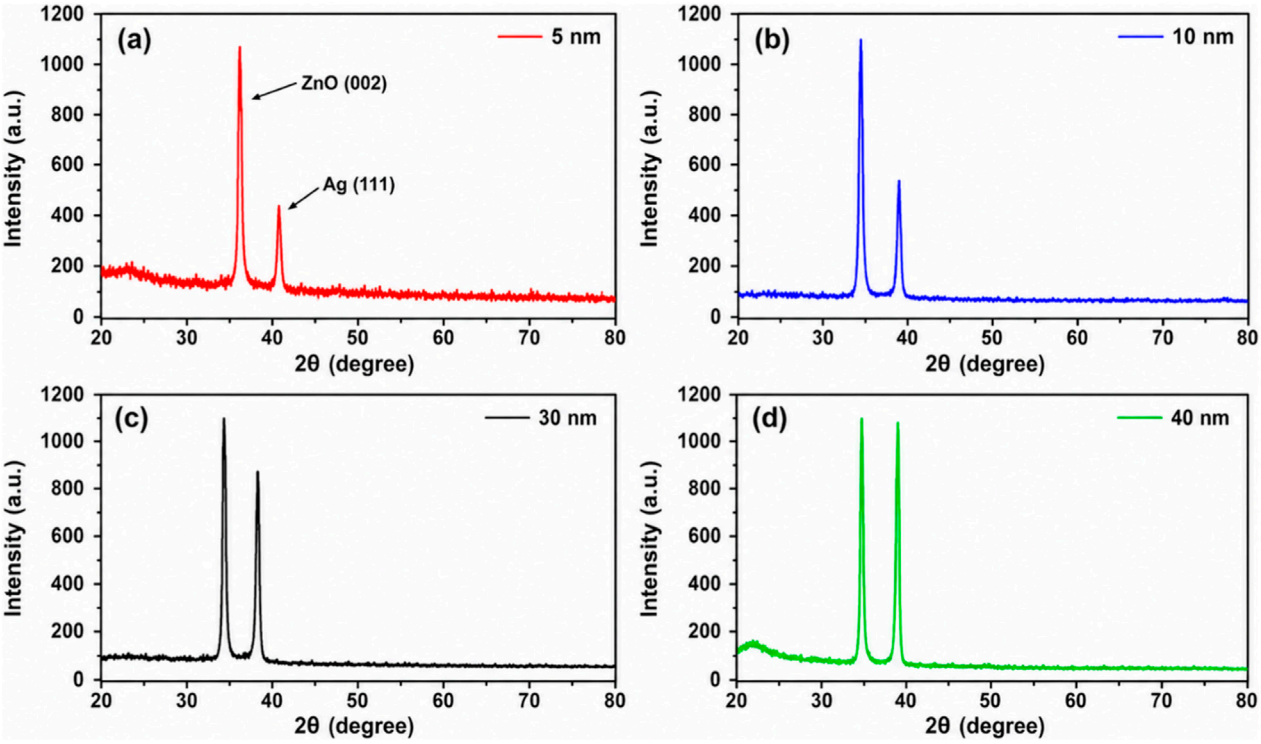


FIGURE 4
X-ray diffraction patterns of ZnO (200 nm)/Ag/ZnO (50 nm) multilayer thin films with varying Ag layer thicknesses: **(a)** 5 nm, **(b)** 10 nm, **(c)** 30 nm, and **(d)** 40 nm (Thabit et al., 2021).

plasmonic absorption associated with metallic interlayers, defect-induced absorption states, and dielectric interference effects. These optical mechanisms directly determine the suitability of multilayer structures for transparent and conductive applications (Kong and Lee, 2020).

Multilayer structures typically exhibit high transmittance (~80–90%) in the visible region, with absorption occurring primarily in the ultraviolet region. However, changes in the metal thickness or defect concentration can significantly alter the optical behaviour. For instance, an increased ion fluence has been shown to reduce the transmittance due to increased scattering and defect-induced absorption (Li et al., 2021).

The optical bandgap is commonly determined using Tauc plots, which provide insights into the electronic transitions and defect states. In TiO_2-based multilayers, ion implantation has been observed to reduce the bandgap from ~2.8 eV to ~2.7 eV, indicating the formation of defect states that enhance electrical conductivity.

Additionally, the refractive index of the dielectric layers plays a crucial role in controlling optical interference and anti-reflective properties. Therefore, UV–Vis spectroscopy is essential for optimizing the optical performance of multilayer structures.

The Tauc relation describes the dependence of optical absorption on photon energy and is widely used to estimate the electronic bandgap by extrapolating the linear portion of the plot. This represents the minimum photon energy required to excite electrons from the valence band to the conduction band. When the incident photon energy (hν) exceeds the optical bandgap (E.g.,), electrons absorb sufficient energy to transition to higher energy states, leading to increased optical absorption. The standard expression was used to compute the optical bandgap associated with the TAT films (Tauc, 1974).

$$\alpha h\vartheta = B(h\vartheta - Eg)^n \tag{3}$$

In Equation 3, hϑ represents the energy of the photon (eV), α denotes the absorption coefficient (cm^{-1}), E.g., is the optical band gap (eV), B is a material constant, and n is the transition exponent, where n = 2 for indirect allowed transitions and n = ½ for direct allowed transitions.

TiO_2 exhibits an indirect bandgap transition mechanism, which is explained by taking n = 2. The absorption coefficient (α) was determined using Beer-Lambert's law (Joseph et al., 2006). This equation quantifies the attenuation of light as it propagates through a multilayer structure. It describes the strength with which a material absorbs incident electromagnetic radiation per unit thickness. A larger absorption coefficient indicates stronger photon absorption.

$$\alpha = \frac{1}{d}\ln\frac{1}{T} = \frac{2.303}{d}A \tag{4}$$

In Equation 4, A is the absorbance of the TAT film, d is the thickness of the multilayer structure (nm), and T is the transmittance. Thus, combining the Tauc relation with Beer–Lambert analysis enables a comprehensive

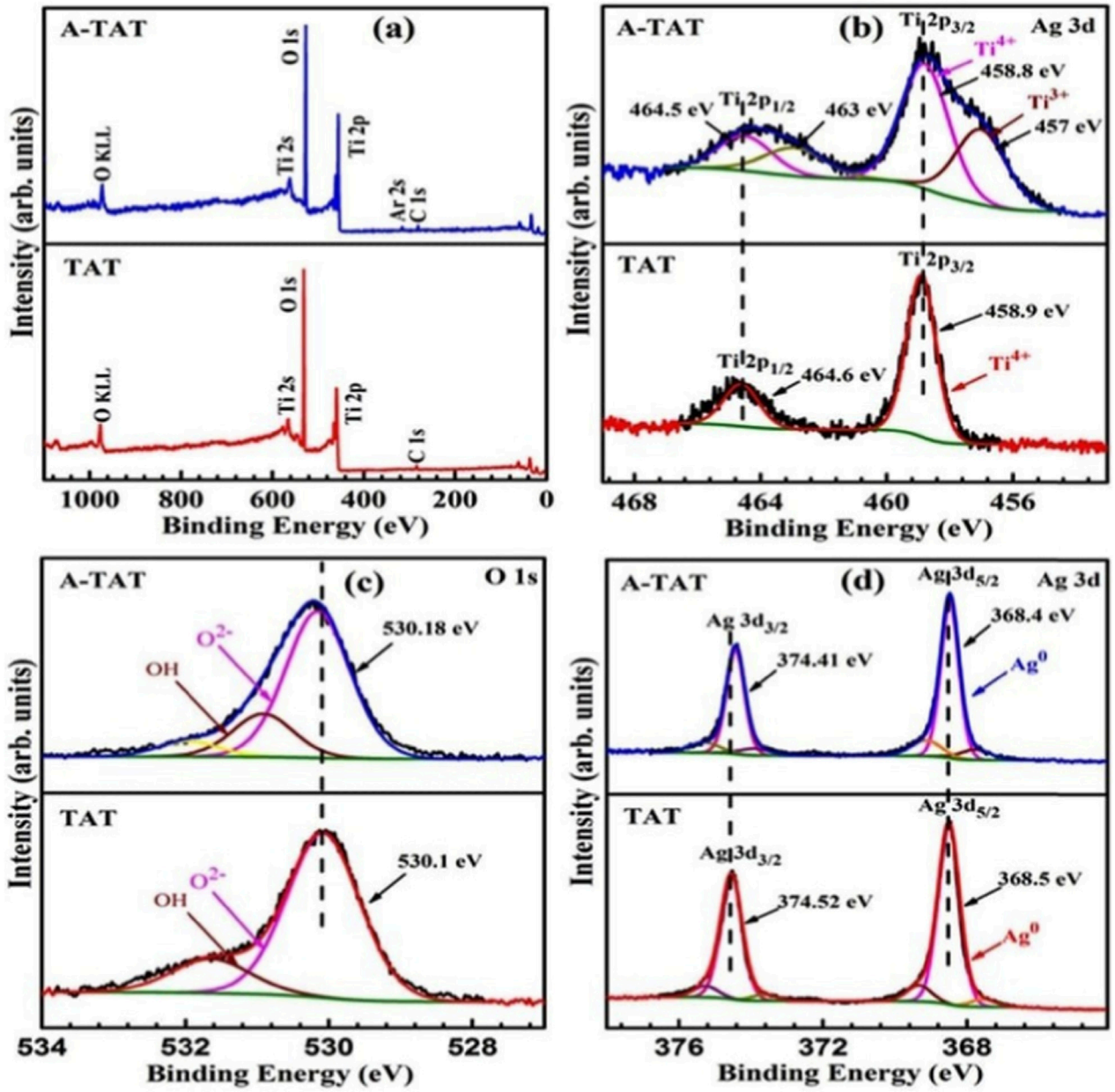


FIGURE 5
XPS spectra of pristine and Ar-implanted (1 × 10^{16} ions/cm^2) TiO_2/Ag/TiO_2 multilayer films: **(a)** survey scan, **(b)** Ti 2p, **(c)** O 1s, and **(d)** Ag 3days (Singh et al., 2018).

understanding of the optical behaviour, electronic structure, and transparency-performance trade-off in multilayer optoelectronic systems.

Figure 7a shows that the optical transmittance decreases slightly after implantation owing to increased defect-induced scattering. Figure 7b shows the corresponding increase in absorbance. The Tauc plots in Figure 7c indicate a small reduction in optical bandgap, while Figure 7d demonstrates only minor variation in refractive index, confirming that implantation modifies the electronic structure without significantly degrading optical performance (Singh et al., 2017).

4.5 Hall measurements: charge transport mechanism

Hall effect measurements are critical for evaluating the electrical properties of multilayer structures, including carrier concentration, mobility, resistivity, and sheet resistance of the layers. These parameters are essential for assessing the suitability of materials as transparent conducting electrodes.

In multilayer systems, the electrical conductivity is primarily governed by the metallic layer, whereas the dielectric layers influence carrier scattering and interface resistance. Hall measurements

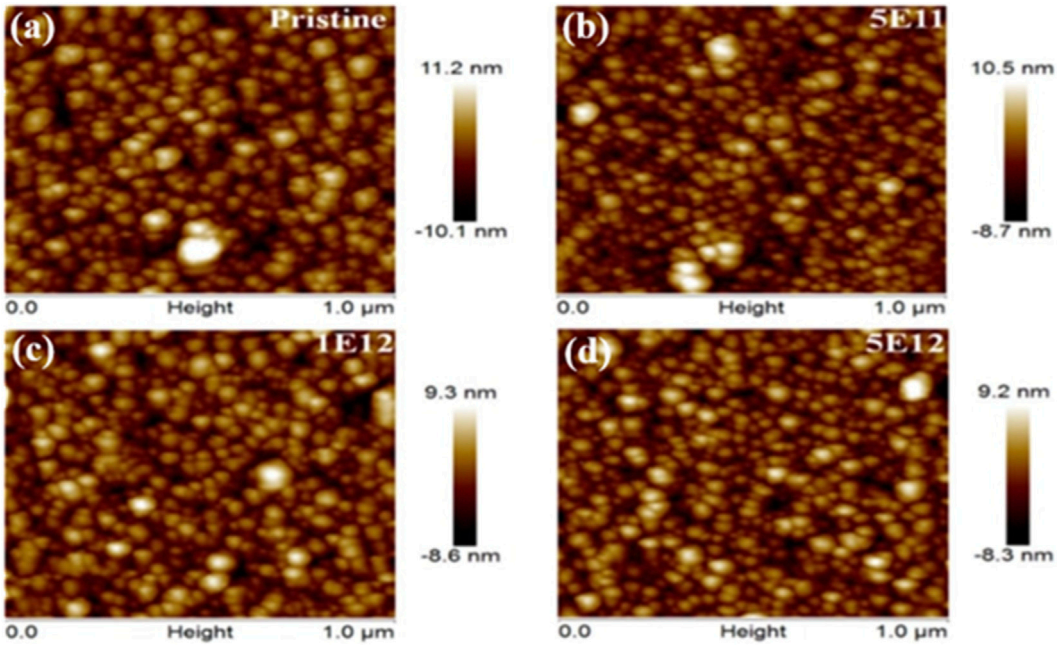


FIGURE 6
AFM surface morphology of ZAZ multilayer structures: **(a)** pristine film, **(b)** irradiated at 5 × 10^{11} ions cm^{-2}, **(c)** irradiated at 1 × 10^{12} ions cm^{-2}, and **(d)** irradiated at 5 × 10^{12} ions cm^{-2} (Sharma et al., 2017).

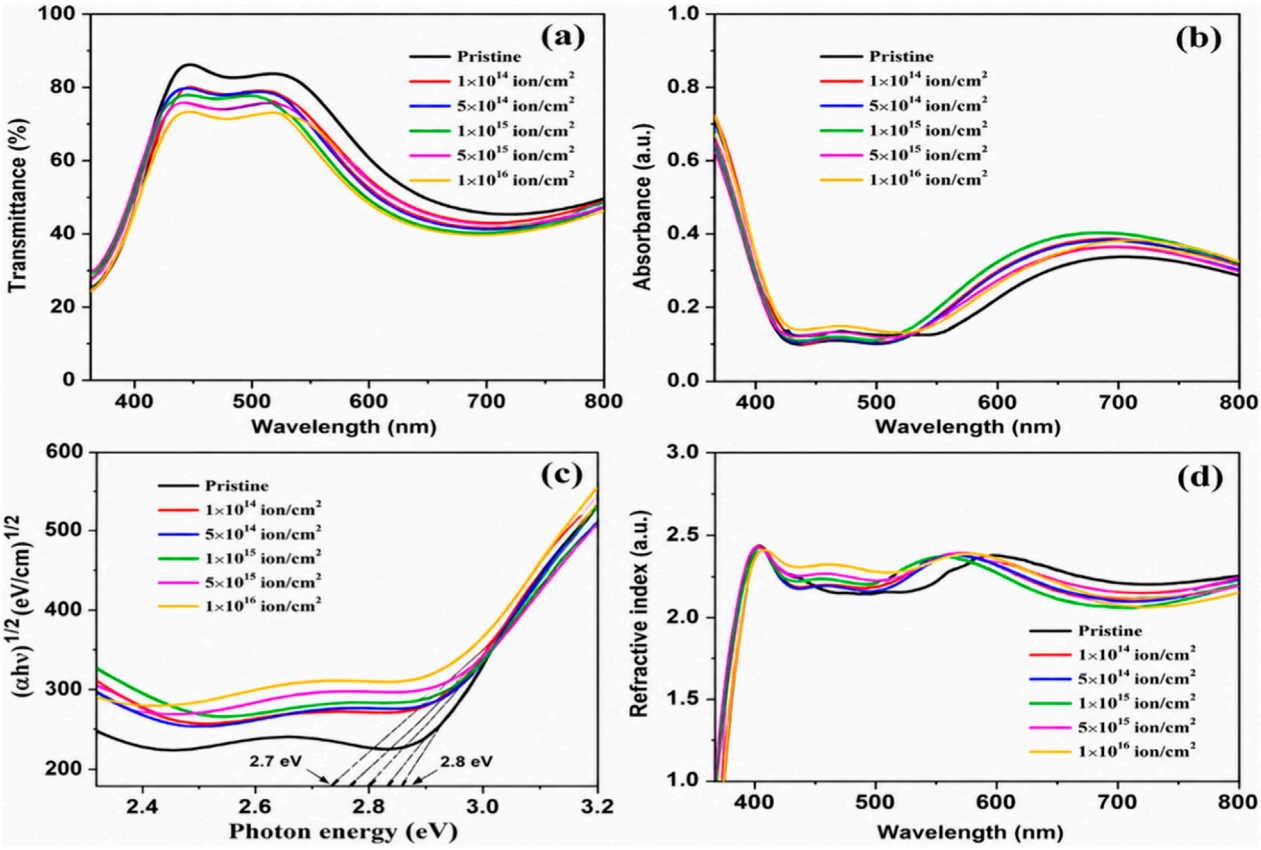


FIGURE 7
(a) Transmittance, **(b)** absorbance, **(c)** Tauc plot, and **(d)** refractive index for both pristine and implanted films of TAT (Singh et al., 2017).

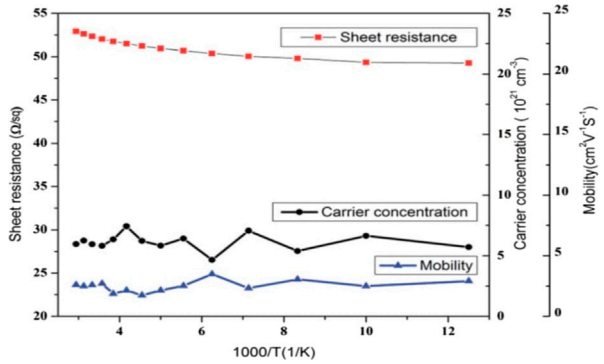


FIGURE 8
Reciprocal temperature dependence of the carrier concentration, mobility, and sheet resistance of the SnOx/Au/SnOx multilayer thin film (Sharma et al., 2016b).

provide insights into how structural parameters, such as metal thickness and interface quality, affect charge transport.

The Van der Pauw method is commonly used for thin films because of its accuracy and flexibility in the sample geometry. A key observation in multilayer structures is that an optimized metal thickness and smooth interfaces lead to higher carrier mobility and lower resistivity, which are essential for high-performance devices.

Figure 8 illustrates how the carrier concentration, mobility, and sheet resistance of the SnO_x/Au/SnO_x multilayers change with temperature. The carrier concentration and mobility remained relatively stable over the investigated temperature range, whereas the sheet resistance exhibited only slight variation, demonstrating the excellent thermal stability of the multilayer electrode (Sharma et al., 2016b).

4.6 Raman spectroscopy: vibrational properties and structural disorder

Raman spectroscopy, a technique that uses laser light to analyse materials, was used to study the vibrational modes and phonon interactions in multilayer structures. This provides valuable information on the crystallinity, phase composition, and defect states (Shen et al., 2022).

In oxide-based multilayers such as SnO_2, characteristic REq.es (A_1g, B_2g, and, E.g.,) can be used to confirm phase purity and structural integrity (Diéguez et al., 2001). The broadening of the Raman peaks often indicates a reduced crystallite size or increased disorder, which can arise from ion irradiation or interface strain.

Additionally, Raman spectroscopy can detect contributions from substrates (e.g., SiO_2), which is particularly relevant for highly transparent, thin films. Thus, Raman analysis complements XRD and XPS in providing a comprehensive understanding of the structural properties. Figure 9 shows the Raman spectrum of the SnO_x/Au/SnO_x multilayer. The characteristic Raman modes confirm the preservation of the oxide crystal structure after the fabrication of the multilayers. The absence of significant peak broadening or additional impurity peaks indicates minimal structural degradation and good interface quality (Sharma et al., 2016b).

4.7 Structure–property–performance correlation

A key takeaway from the characterization studies is that the performance of metal–dielectric multilayer structures is governed by a complex interaction between the structural quality, interface chemistry, and defect states. No single characterization technique is sufficient; rather, a multi-technique approach is required to fully understand and optimize these systems.

- Characterization is essential for understanding the structure–property relationships.
- The interface quality and defect states critically influence the performance.
- Multilayer systems require the use of combined analytical techniques.
- The optimization of the structural, optical, and electrical properties must be performed simultaneously.

The performance of D/M/D multilayer transparent conductive systems emerges from the coupled interactions among fabrication methodology, structural quality, interface chemistry, and optoelectronic transport mechanisms. High electrical conductivity requires continuous metallic pathways, whereas high optical transparency requires the minimization of metallic absorption and scattering losses. These competing requirements make multilayer optimization fundamentally dependent on precise thickness control, interface smoothness, and defect engineering. Characterization techniques such as XRD, XPS, AFM, and UV-visible spectroscopy collectively provide complementary insights into these interconnected performance determinants. Therefore, an effective multilayer design requires the integrated consideration of processing conditions, structural characteristics, and application-specific performance requirements, rather than the isolated optimization of individual parameters (Dhar and Alford, 2012).

5 Emerging trends and future perspectives in multilayer transparent optoelectronic systems

Although dielectric/metal/dielectric (D/M/D) multilayer transparent conductive structures have demonstrated substantial progress as transparent electrode platforms, contemporary developments in optoelectronics indicate that future technological relevance will increasingly depend on multifunctionality, adaptive interfaces, and device-level integration rather than solely on conventional transparency–conductivity optimization of the structures. The emerging landscape of optoelectronic materials increasingly emphasizes interface-controlled functionality, hybrid architectures, bio-integrated systems, and application-specific device engineering (Sengupta et al., 2026). Consequently, the future of D/M/D multilayers should be considered within a broader interdisciplinary framework.

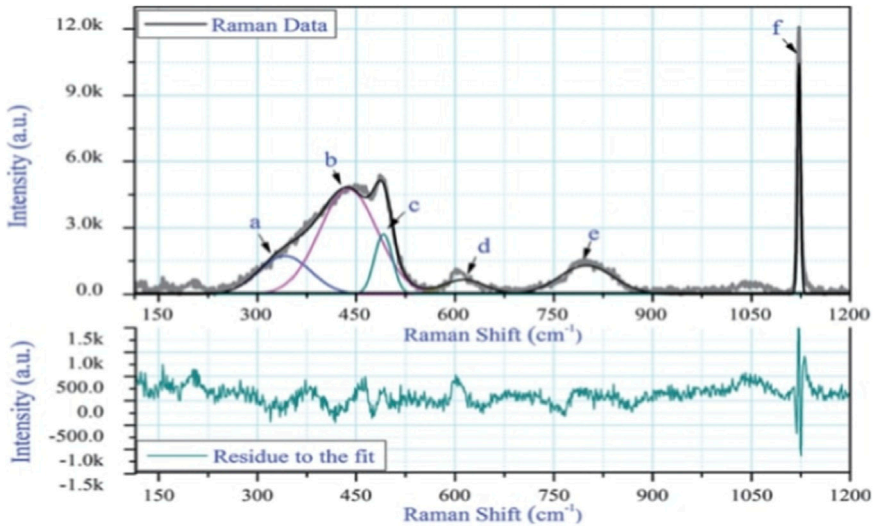


FIGURE 9
Raman spectrum of SnO_x/Au/SnO_x stacked multilayer (Sharma et al., 2016b).

5.1 Interface-driven functionality beyond conventional transparent conductivity

Traditional multilayer transparent conductive research primarily focuses on minimizing the sheet resistance while preserving the visible optical transparency. However, modern optoelectronic systems increasingly rely on interface-driven functionality, where interfaces actively determine charge extraction, energy level alignment, recombination suppression, defect passivation, and operational stability (Sharma et al., 2026).

In multilayer optoelectronic architectures, dielectric/metal interfaces are no longer passive structural boundaries but electronically active regions that critically influence carrier transport and device efficiency. Interface roughness, chemical interdiffusion, defect-state formation, and local electric field redistribution can substantially alter the charge mobility, contact resistance, and optical losses (Yao Z. et al., 2024). Therefore, future multilayer designs must move beyond thickness optimization toward deliberate interface engineering.

Recent device-level studies have demonstrated that atomically controlled interfaces can significantly improve charge extraction and spectral selectivity in advanced optoelectronic systems, particularly in tandem photovoltaic architectures, where transparent conductive electrodes must simultaneously satisfy conductivity, transparency, and energy-alignment constraints (Cha et al., 2025). This evolution indicates a transition from passive multilayer conductors to interface-functional optoelectronic platforms.

Recent experimental studies have demonstrated that interface engineering plays a decisive role in improving the performance of multilayered transparent electrodes. For example, interface-modified ZnO/Ag/ZnO and Nb_2O_5/Ag/Nb_2O_5 multilayer electrodes exhibited enhanced carrier extraction and reduced interfacial recombination owing to improved dielectric/metal contact quality. Similarly, interface-controlled transparent electrodes have been successfully employed in tandem perovskite–silicon solar cells, where precise energy-level alignment and defect passivation significantly improved the power conversion efficiency while maintaining high optical transparency (Yao Z.-F. et al., 2024; Cha et al., 2025; Sharma et al., 2026). These examples demonstrate that future D/M/D multilayers should be designed not only for low sheet resistance but also for interface-controlled electronic functionality.

5.2 Device-level integration and system-oriented design

A major limitation of earlier multilayer research has been the predominant emphasis on isolated material performance metrics, such as transmittance, resistivity, and thickness optimization, often without sufficient consideration of complete device integration.

In practical optoelectronic applications, material-level optimization alone is insufficient for practical optoelectronic applications. Device performance depends simultaneously on multiple interconnected parameters, including.

- charge injection efficiency,
- contact resistance,
- optical spectral management,
- environmental durability,
- thermal stability,
- mechanical reliability,
- manufacturing compatibility.

Consequently, contemporary research increasingly evaluates transparent conductive multilayers within integrated device systems, including tandem solar cells, transparent heaters, photodetectors, OLEDs, smart coatings, and flexible electronic platforms (Wang et al., 2021; Sayem et al., 2022; Yao Z. et al., 2024).

For example, recent multilayer transparent electrode architectures developed for tandem perovskite–silicon

photovoltaic systems demonstrate that spectral engineering and interface optimization are equally critical as conventional conductivity metrics (Li and Zhang, 2020). Such studies emphasize that the future competitiveness of D/M/D multilayers will depend increasingly on system-level functionality rather than isolated material performance.

Examples based on real-world devices further support the need for systematic system optimization. ZnO/Ag/ZnO and ITO/Ag/ITO multilayer electrodes have been effectively used in transparent heaters, OLEDs, touch screens, and flexible photovoltaic systems because of their excellent balance between optical transparency and electrical conductivity (Wang et al., 2021; Yao Z. et al., 2024). Similarly, multilayers of Ta_2O_5/Ag/Ta_2O_5 and Nb_2O_5/Ag/Nb_2O_5 were shown to be efficient transparent electrodes for perovskite and tandem solar cells, where spectrum management and interface engineering optimization resulted in increased carrier collection and improved device stability (Li and Zhang, 2020; Sayem et al., 2022). The findings clearly indicate that future multilayer development must focus on the overall device performance rather than on the properties of a single material.

5.3 Hybrid organic–inorganic and biohybrid optoelectronic platforms

One of the most transformative emerging directions in optoelectronics involves hybrid architectures integrating inorganic conductive systems with soft organic, biomolecular, or biologically interactive functional materials (Sharfstein, 2022; Zhou et al., 2024).

Recent high-impact studies have demonstrated that optoelectronic interfaces can be engineered through structurally ordered hybrid heterointerfaces exhibiting adaptive charge transport, programmable optoelectronic behaviour, and biologically relevant functionality (Kuang et al., 2026). For example, the lattice-guided assembly of optoelectronically active π-conjugated peptide systems on one-dimensional van der Waals substrates has been shown to enhance photocurrent generation through the formation of structurally ordered hybrid interfaces. These systems represent a fundamentally different design paradigm compared to conventional rigid transparent conductors.

Similarly, emerging biohybrid materials increasingly combine electronic transport with biological compatibility, dynamic interface responsiveness, and multifunctional signal transduction (Wang et al., 2025).

Compared with such emerging platforms, conventional D/M/D multilayers retain several important advantages.

- superior electrical conductivity,
- excellent thermal robustness,
- mature vacuum-process compatibility,
- well-established industrial manufacturability,
- stable optical transparencies.

However, their limitations become increasingly apparent in specialized applications that require the following:

- mechanical softness,
- conformability,
- biological integration,
- adaptive interfacial response,
- multifunctional-sensing behaviour.

Thus, while D/M/D multilayers remain highly effective for traditional transparent conductive applications, hybrid bio-integrated systems may increasingly define future application-specific optoelectronic interfaces in the future.

Several studies have demonstrated the practical implementation of hybrid and biohybrid optoelectronic systems. Likewise, recent Cell Biomaterials research has demonstrated biohybrid electronic platforms capable of establishing efficient electronic communication between synthetic materials and living tissues, thereby enabling advanced biosensing, tissue engineering, and implantable bioelectronic devices with excellent biological compatibility (Zhou et al., 2024). Furthermore, adaptive hybrid optoelectronic interfaces reported in Advanced Materials have shown dynamically tunable electronic responses by combining inorganic conductive components with soft organic materials, opening new opportunities for multifunctional photodetectors and wearable optoelectronic systems (Sharfstein, 2022). These representative examples clearly illustrate that hybrid organic–inorganic architectures extend transparent electronics beyond conventional transparent electrodes toward intelligent, adaptive, and biologically integrated optoelectronic systems, while complementing the high conductivity and thermal stability offered by conventional D/M/D multilayers (Wang et al., 2025).

5.4 Polymeric transparent conductive systems: comparative perspective

A comparison between metal-based multilayer systems and polymeric transparent conductive platforms provides an additional perspective on emerging technological directions.

Polymeric conductive materials—including PEDOT: PSS-based electrodes, stretchable conductive polymers, hydrogel conductors, and polymer-nanocomposite transparent systems—offer distinct advantages in applications requiring lightweight, mechanically compliant, and low-temperature processable conductive interfaces (Khan et al., 2023; Dallaev, 2025).

The key advantages of polymeric systems include:

- exceptional mechanical flexibility,
- stretchability,
- lightweight construction,
- conformal surface integration,
- compatibility with wearable electronics,
- suitability for biological interfaces.

However, compared with D/M/D multilayer structures, polymeric systems often exhibit limitations, such as:

- lower electrical conductivity,
- moisture sensitivity,
- reduced thermal stability,
- long-term environmental degradation,
- lower structural robustness.

Thus, the suitability of application is strongly context-dependent.

For conventional optoelectronic technologies, such as.

- rigid displays,
- photovoltaic devices,
- transparent heaters,
- architectural coatings,

D/M/D multilayers are technologically superior.

In contrast, for:

- wearable bioelectronics,
- adaptive sensors,
- soft human–machine interfaces,
- implantable optoelectronic systems,

Polymeric and hybrid platforms may offer more suitable performance characteristics (Aljabali et al., 2025).

This broader comparative framework enhances the technological relevance of D/M/D multilayer research by positioning it within an evolving transparent electronics ecosystem.

Numerous polymeric transparent conductive systems have been demonstrated in practical devices. PEDOT: PSS has been extensively employed as a transparent electrode in flexible organic solar cells, OLEDs, and touch-sensitive electronic devices because of its excellent transparency, low-temperature solution processability, and mechanical flexibility (Dallaev, 2025). Conductive hydrogels have recently demonstrated remarkable performance in wearable strain sensors, electronic skin, and soft robotic systems by providing simultaneous electrical conductivity, stretchability, and self-healing capabilities (Khan et al., 2023). Similarly, silver nanowire/PEDOT: PSS composite electrodes have been successfully utilized in flexible transparent heaters and foldable displays, where the polymer matrix improves adhesion and mechanical robustness, while silver nanowires maintain excellent electrical conductivity (Aljabali et al., 2025). Although these polymeric systems outperform conventional transparent electrodes in terms of flexibility and biocompatibility, D/M/D multilayers continue to exhibit superior electrical conductivity, thermal stability, and environmental durability, making them the preferred choice for photovoltaic modules, transparent heaters, display technologies, and architectural glazing applications (Dalapati et al., 2021).

Figure 10 is the representation of the comparison of D/M/D, Polymeric, and Biohybrid Transparent Conductive Platforms for different optoelectronics applications.

5.5 AI-assisted and computational multilayer design

Another emerging direction involves the transition from empirical multilayer optimization to predictive computational design.

Historically, multilayer performance optimization has relied heavily on experimental trial-and-error variations in thickness, deposition conditions, and material combinations. However, future multilayer engineering is increasingly likely to benefit from predictive methodologies that integrate the following:

- electromagnetic optical modelling,
- density functional theory calculations,
- inverse multilayer design algorithms,
- machine-learning-guided material discovery.

These computational approaches can accelerate the optimization of:

- metallic thickness,
- dielectric refractive index,
- interface energetics,
- plasmonic coupling,
- spectral selectivity,
- defect-tolerant architecture.

AI-assisted inverse design may substantially reduce experimental iterations and enable application-specific multilayer engineering (Yao et al., 2025).

Practical examples of computational material design have already demonstrated the effectiveness of artificial intelligence in transparent electrode optimization. Recent machine learning algorithms have been employed to predict the optimum dielectric and metallic layer thicknesses that maximize the Haacke figure of merit while simultaneously minimizing optical reflection and sheet resistance. Density functional theory (DFT) calculations have also been widely used to investigate the interface energetics, electronic band alignment, oxygen-vacancy formation, and charge transfer across oxide/metal interfaces prior to experimental fabrication. Furthermore, electromagnetic transfer-matrix modelling has enabled the rapid optimization of multilayer interference effects in transparent electrodes for tandem perovskite–silicon solar cells, smart windows, and transparent heaters by accurately predicting the wavelength-dependent optical transmission and reflection characteristics (Yao et al., 2025). These computational approaches considerably reduce experimental trial-and-error and are expected to become indispensable tools for designing next-generation D/M/D multilayer architectures.

5.6 Sustainable manufacturing and commercial translation

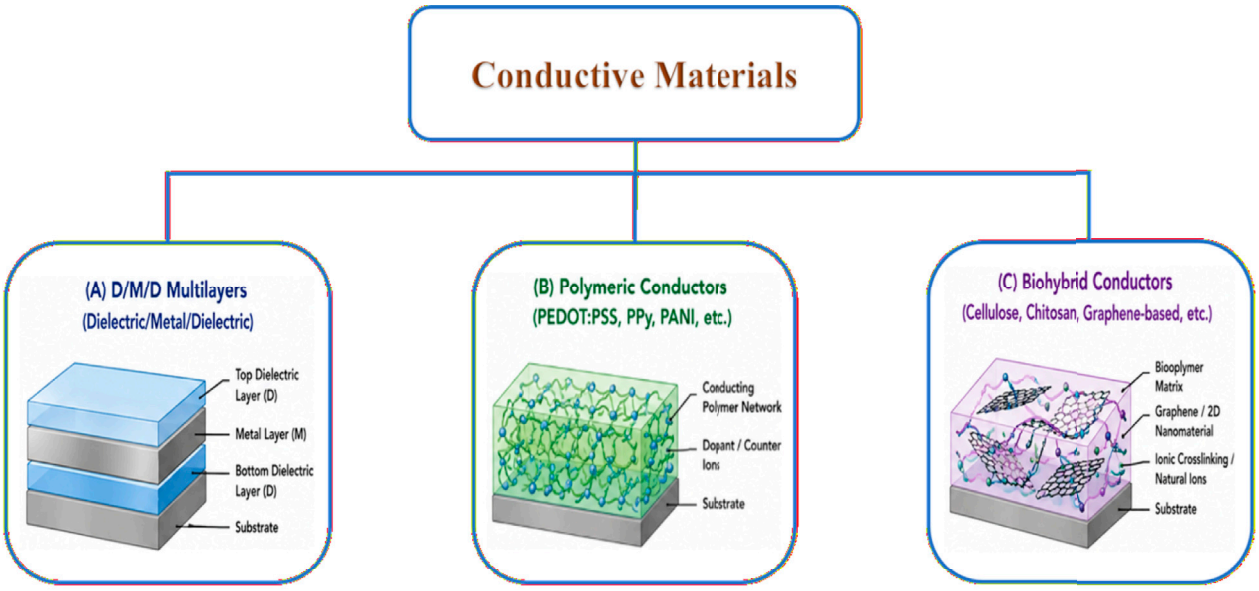
Commercial relevance increasingly depends not only on the optoelectronic performance but also on the sustainability of the manufacturing process and economic feasibility.

The important priorities include:

- reduced dependence on noble metals,
- Cu-protected or alternative metallic interlayers,
- low-energy deposition routes,
- roll-to-roll scalable architectures,
- recyclable substrates,
- environmentally sustainable fabrication strategies for them.

Economic sustainability may become equally important as conductivity and transparency in determining industrial adoption.

Several recent investigations have demonstrated promising pathways toward the large-scale commercialization of D/M/D multilayer transparent electrodes. Roll-to-roll magnetron sputtering



Performance Criteria	D/M/D Multilayers (Dielectric/Metal/Dielectric)	Polymeric Conductors (PEDOT:PSS, PANI, etc.)	Biohybrid Conductors (Cellulose, Chitosan, etc.)	Key Strengths
Electric Conductivity (Low Sheet Resistance)	★★★★★ (10^{-6}–10^{-4} S/cm)	★★★☆☆ (10^{-4}–10^{-2} S/cm)	★★★☆☆ (10^{-5}–10^{-3} S/cm)	High conductivity with excellent transparency
Optical Transparency (Visible Range)	★★★★★ (>90%)	★★★★☆ (80–90%)	★★★★☆ (70–90%)	Broad visible transparency
Flexibility/Mechanical Compliance	★★★☆☆ (Moderate)	★★★★★ (Excellent)	★★★★☆ (Good to Excellent)	Excellent flexibility for wearable/flexible devices
Thermal/Environmental Stability	★★★★★ (Excellent)	★★☆☆☆ (Limited)	★★★☆☆ (Sensitive to moisture and temperature)	Superior stability under harsh conditions
Biocompatibility/Biointegration	★★☆☆☆ (Low to Moderate)	★★★☆☆ (Good)	★★★★★ (Excellent)	Low toxicity; suitable for bioelectronics
Manufacturability/Scalability	★★★★☆ (High – Sputtering, ALD, etc.)	★★★☆☆ (Moderate – Solution based)	★★☆☆☆ (Challenging – Consistency and scale-up)	Compatible with large-area and roll-to-roll processes
Cost & Material Abundance	★★★☆☆ (Moderate – Ag is costly)	★★☆☆☆ (Low to Moderate)	★★★★☆ (Low – Abundant natural resources)	Utilizes earth-abundant and low-cost materials
Technology Readiness (Industrial Maturity)	★★★★☆ (High – Commercially established)	★★★☆☆ (Moderate – Emerging)	★★☆☆☆ (Low to Moderate – Early stage)	Closest to commercial deployment
Rating Scale: ★★★★★ Excellent, ★★★★☆ Good, ★★★☆☆ Moderate, ★★☆☆☆ Limited, ★☆☆☆☆ Poor				

FIGURE 10
Comparison of transparent conductive platforms for optoelectronic applications.

has successfully produced ZnO/Ag/ZnO and ITO/Ag/ITO multilayer electrodes on flexible polyethylene terephthalate (PET) substrates with excellent thickness uniformity, optical transparency, and production throughput suitable for industrial manufacturing (Wang et al., 2021; Arun et al., 2025). To reduce manufacturing costs, researchers have investigated Cu-protected multilayers, Ag–Cu alloy interlayers, and oxide/metal hybrid structures as economically attractive alternatives to pure Ag-based electrodes while maintaining competitive optoelectronic performance (Pastuszak and Węgierek, 2022). Moreover, low-temperature vacuum deposition compatible with flexible polymer substrates has enabled the energy-efficient fabrication of transparent electrodes for wearable electronics and flexible displays (Sharma et al., 2016a). The adoption of recyclable polymer substrates, environmentally benign sputtering targets, and low energy deposition techniques further supports the sustainable manufacturing. These representative examples demonstrate that the future commercial success of D/M/D multilayers will depend not only on superior optical and electrical performance but also on scalable manufacturing, reduced material consumption, environmental sustainability, and cost-effective production strategies.

5.7 Strategic outlook

The future of transparent optoelectronics will likely be defined not by a single dominant material class but by

the convergence of multiple technology platforms, including inorganic multilayers, polymeric transparent conductors, hybrid organic–inorganic systems, and bioelectronic optoelectronic architectures (Nallusamy et al., 2025).

Within this broader technological landscape, D/M/D multilayers remain highly competitive for applications that demand exceptional conductivity, optical transparency, and manufacturing maturity. However, their long-term relevance increasingly depends on adaptation toward multifunctionality, intelligent interface engineering, scalable sustainability, and compatibility with emerging hybrid device paradigms.

6 Conclusion

Dielectric/metal/dielectric (D/M/D) multilayer transparent conductive structures represent one of the most promising material platforms for overcoming the intrinsic limitations of conventional transparent conducting oxides (TCOs) in advanced optoelectronic applications. This review critically examines the structural design principles, fabrication methodologies, characterization strategies, performance optimization approaches, emerging applications, and industrial scalability challenges associated with these multilayer systems.

The analysis demonstrates that the superior performance of the D/M/D architecture arises from the synergistic interaction between the conductive metallic interlayers and optically transparent dielectric coatings, allowing the simultaneous optimization of electrical conductivity and optical transparency beyond the limitations of conventional single-layer electrodes. Comparative analysis of representative systems confirms that Ag-based multilayers, such as ZnO/Ag/ZnO, ITO/Ag/ITO, and Ta_2O_5/Ag/Ta_2O_5, currently provide some of the most favorable combinations of high optical transmittance (>90%) and low sheet resistance (<5 Ω/sq), making them highly attractive for transparent electrode applications.

Despite these encouraging advances, several important scientific and technological challenges remain unresolved. A major limitation is the long-term stability of the ultrathin metallic interlayers. Although silver offers exceptional conductivity and favorable optical characteristics, it remains susceptible to oxidation, agglomeration, thermal diffusion, and interfacial degradation, which can compromise device reliability under the operational conditions. Gold-based multilayers provide improved chemical stability; however, their significantly higher material cost limits their industrial applicability.

The scalability of fabrication remains a critical barrier. Magnetron sputtering has emerged as the dominant deposition method because of its excellent control over film thickness, interface quality, and reproducibility. However, vacuum-based fabrication remains capital-intensive, energy-demanding, and economically challenging for large-area production. Maintaining nanometer-scale thickness precision, particularly for ultrathin metallic layers, across industrial substrate dimensions remains technically challenging, as even minor thickness deviations can substantially alter optical and electrical performance.

This review also highlights that interface engineering remains one of the most decisive factors governing multilayer performance. Interface roughness, chemical interdiffusion, defect accumulation, and contamination strongly affect carrier mobility, optical scattering, adhesion, and the long-term structural stability of these devices. Consequently, a deeper mechanistic understanding of the interfacial electronic structure remains essential.

Defect engineering has emerged as an effective strategy for enhancing the functionality of multilayers. Controlled ion irradiation, vacancy engineering, and interface-state modification have demonstrated the ability to tailor carrier transport, optical bandgap, and plasmonic response beyond conventional thickness optimization techniques. However, excessive defect generation may degrade the transparency and structural integrity of the material, requiring careful optimization.

From an application perspective, D/M/D multilayer structures are rapidly evolving beyond traditional transparent conductive electrodes. Recent progress has demonstrated their applicability in flexible electronics, photodetectors, semitransparent photovoltaics, transparent heat mirrors, plasmonic sensing, and wearable optoelectronic systems. Their growing integration into emerging tandem photovoltaic architectures further highlights their technological importance.

However, their commercial readiness remains limited. Current research remains strongly focused on laboratory-scale performance optimization, whereas comparatively less attention has been directed toward manufacturing yield, process reproducibility, environmental durability, cost reduction, and sustainable material alternatives.

Future research should prioritize the following:

- low-cost alternatives to noble metal interlayers,
- atomically controlled interface engineering,
- roll-to-roll compatible scalable manufacturing,
- stabilization of ultrathin continuous metallic films,
- hybrid multilayer integration with emerging semiconductor platforms,
- computationally guided multilayer optimization,
- environmentally sustainable processing strategies for them.

Overall, dielectric/metal/dielectric multilayer structures have matured into a scientifically robust and technologically versatile platform for transparent optoelectronic applications. However, their successful transition from high-performance laboratory demonstrations to commercially deployable technologies will require coordinated advances in materials engineering, scalable fabrication, interface stabilization, and economic viability of the devices. With continued innovation, multilayer architectures are expected to play a transformative role in next-generation flexible, energy-efficient, and multifunctional optoelectronic systems.

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SC: Conceptualization, Data curation, Investigation, Methodology, Writing – original draft. AS: Supervision, Validation, Writing – review and editing.

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References

Ahmad, K. S., and Jaffri, S. B. (2018). Phytosynthetic Ag doped ZnO nanoparticles: semiconducting green remediators. *Open Chem.* 16, 556–570. doi: 10.1515/chem-2018-0060

Akhmedov, A., Abduev, A., Kanevsky, V., Muslimov, A., and Asvarov, A. (2020). Low-Temperature fabrication of high-performance and stable GZO/Ag/GZO multilayer structures for transparent electrode applications. *Coatings* 10, 269. doi: 10.3390/coatings10030269

Al-Kuhaili, M. F., Al-Aswad, A. H., Durrani, S. M. A., and Bakhtiari, I. A. (2012). Energy-saving transparent heat mirrors based on tungsten oxide–gold WO3/Au/WO3 multilayer structures. *Sol. Energy* 86, 3183–3189. doi: 10.1016/j.solener.2012.08.008

Ali, H. A., Aziz, I. A.-K., and Abbas, A. S. (2023). "Characterization of Cu-Ag multilayers deposited on brass substrate using electroplating technique," in *AIP Conf. Proc.* doi: 10.1063/5.0164832

Aljabali, A. A. A., Alkaraki, A., Gammoh, O., Qnais, E., Alqudah, A., Mishra, V., et al. (2025). Design, structure, and application of conductive polymer hybrid materials: a comprehensive review of classification, fabrication, and multifunctionality. *RSC Adv.* 15, 27493–27523. doi: 10.1039/D5RA04634C

Arun, A. P., Sreenivasan, N., Patil, J. H., Kusanur, R., Ramachandraiah, H. L., and Ramakrishna, M. (2025). Thin films for next generation technologies: a comprehensive review of fundamentals, growth, deposition strategies, applications, and emerging frontiers. *Processes* 13, 3846. doi: 10.3390/pr13123846

Arya, S., Mahajan, P., Mahajan, S., Khosla, A., Datt, R., Gupta, V., et al. (2021). Review—Influence of processing parameters to control morphology and optical properties of Sol-Gel synthesized ZnO nanoparticles. *ECS J. Solid State Sci. Technol.* 10, 023002. doi: 10.1149/2162-8777/abe095

Bandyopadhyay, A. S., Saenz, G. A., and Kaul, A. B. (2020). Role of metal contacts and effect of annealing in high performance 2D WSe2 field-effect transistors. *Surf. Coat. Technol.* 381, 125084. doi: 10.1016/j.surfcoat.2019.125084

Barman, B., Swami, S. K., and Dutta, V. (2021). Fabrication of highly conducting ZnO/Ag/ZnO and AZO/Ag/AZO transparent conducting oxide layers using RF magnetron sputtering at room temperature. *Mater. Sci. Semicond. process.* 129, 105801. doi: 10.1016/j.mssp.2021.105801

Bokov, D., Turki Jalil, A., Chupradit, S., Suksatan, W., Javed Ansari, M., Shewael, I. H., et al. (2021). Nanomaterial by Sol-Gel method: synthesis and application. *Adv. Mater. Sci. Eng.* 2021, 5102014. doi: 10.1155/2021/5102014

Çetinkaya, Ç., Çokduygulular, E., Güzelçimen, F., and Kınacı, B. (2022). Functional optical design of thickness-optimized transparent conductive dielectric-metal-dielectric plasmonic structure. *Sci. Rep.* 12, 8822. doi: 10.1038/s41598-022-13038-y

Cha, H.-C., Huang, S.-H., Li, C.-F., Tsai, F.-Y., Su, W.-F., and Huang, Y.-C. (2025). Advances and strategies in scalable coating techniques for flexible perovskite solar cells. *Sustain. Energy Fuels* 9, 5962–6006. doi: 10.1039/D5SE00873E

Chavan, G. T., Kim, Y., Khokhar, M. Q., Hussain, S. Q., Cho, E.-C., Yi, J., et al. (2023). A brief review of Transparent Conducting Oxides (TCO): the influence of different deposition techniques on the efficiency of solar cells. *Nanomaterials* 13, 1226. doi: 10.3390/nano13071226

Chen, J., Zhuang, P., Ge, Y., Chu, H., Yao, L., Cao, Y., et al. (2019). Sublimation-Vapor phase pseudomorphic transformation of template-directed MOFs for efficient oxygen evolution reaction. *Adv. Funct. Mater.* 29, 1903875. doi: 10.1002/adfm.201903875

Crespo-Sosa, A., Muñoz, M., Cheang-Wong, J.-C., Oliver, A., Sániger, J. M., and Bañuelos, J. G. (2003). High energy ion irradiation induced surface roughening in Ag and Cu films. *Appl. Surf. Sci.* 206, 178–186. doi: 10.1016/S0169-4332(02)01213-8

Dalapati, G. K., Sharma, H., Guchhait, A., Chakrabarty, N., Bamola, P., Liu, Q., et al. (2021). Tin oxide for optoelectronic, photovoltaic and energy storage devices: a review. *J. Mater. Chem. A Mater* 9, 16621–16684. doi: 10.1039/D1TA01291F

Dallaev, R. (2025). Conductive polymer thin films for energy storage and conversion: supercapacitors, batteries, and solar cells. *Polym. (Basel)* 17, 2346. doi: 10.3390/polym17172346

Dhar, A., and Alford, T. L. (2012). Optimization of Nb2O5/Ag/Nb2O5 multilayers as transparent composite electrode on flexible substrate with high figure of merit. *J. Appl. Phys.* 112, 103113. doi: 10.1063/1.4767662

Diéguez, A., Romano-Rodríguez, A., Vilà, A., and Morante, J. R. (2001). The complete Raman spectrum of nanometric SnO2 particles. *J. Appl. Phys.* 90, 1550–1557. doi: 10.1063/1.1385573

Dong, Q., Li, H.-L., Zhu, Z.-M., Kong, D.-P., Zhong, S.-L., and Dang, Z.-M. (2025). TiO_x electrode interface modification layer promises the prominent electrical/macroscopic properties of the metallized dielectric films. *RSC Adv.* 15, 6700–6707. doi: 10.1039/D5RA00206K

Girtan, M. (2012). Comparison of ITO/metal/ITO and ZnO/metal/ZnO characteristics as transparent electrodes for third generation solar cells. *Sol. Energy Mater. Sol. Cells* 100, 153–161. doi: 10.1016/j.solmat.2012.01.007

Ho Kim, J., Hwan Lee, J., Kim, S.-W., Yoo, Y.-Z., and Seong, T.-Y. (2015). Highly flexible ZnO/Ag/ZnO conducting electrode for organic photonic devices. *Ceram. Int.* 41, 7146–7150. doi: 10.1016/j.ceramint.2015.02.031

Hong, J., Wang, M., Jiang, J., Zheng, P., Zheng, H., Zheng, L., et al. (2020). Optoelectronic performance of multilayer WSe_2 transistors enhanced by defect engineering. *Appl. Phys. Express* 13, 061004. doi: 10.35848/1882-0786/ab8f13

Jenifer, K., and Parthiban, S. (2023). Highly stable, ultra-thin Au embedded zinc tin oxide multilayer transparent conductive thin films. *Curr. Appl. Phys.* 53, 94–103. doi: 10.1016/j.cap.2023.07.001

Jeon, K., Youn, H., Kim, S., Shin, S., and Yang, M. (2012). Fabrication and characterization of WO3/Ag/WO3 multilayer transparent anode with solution-processed WO3 for polymer light-emitting diodes. *Nanoscale Res. Lett.* 7, 253. doi: 10.1186/1556-276X-7-253

Joseph, B., Manoj, P. K., and Vaidyan, V. K. (2006). Studies on the structural, electrical and optical properties of Al-doped ZnO thin films prepared by chemical spray deposition. *Ceram. Int.* 32, 487–493. doi: 10.1016/j.ceramint.2005.03.029

Kaya, C., Akaoglu, B., Er, U., and Bayram, Y. (2024). Transparent dielectric/metal/dielectric multilayer microstrip patch antennas for X and Ku-band. *Opt. Commun.* 564, 130620. doi: 10.1016/j.optcom.2024.130620

Khan, S., Ali, S., Khan, A., and Bermak, A. (2023). Wearable printed temperature sensors: short review on latest advances for biomedical applications. *IEEE Rev. Biomed. Eng.* 16, 152–170. doi: 10.1109/RBME.2021.3121480

Kong, H., and Lee, H.-Y. (2020). High performance flexible transparent conductive electrode based on ZnO/AgOx/ZnO multilayer. *Thin Solid Films* 696, 137759. doi: 10.1016/j.tsf.2019.137759

Kuang, Y., Yao, Z.-F., Salgado, C., Lundqvist, E. M., Morsi, N., Celt, N., et al. (2026). Optoelectronic biohybrid platform enables light-controlled cardiac structural and functional feedback. *Cell Biomater.*, 100416. doi: 10.1016/j.celbio.2026.100416

Li, H., and Zhang, W. (2020). Perovskite tandem solar cells: from fundamentals to commercial deployment. *Chem. Rev.* 120, 9835–9950. doi: 10.1021/acs.chemrev.9b00780

Li, M., Liu, W., Zhang, F., Zhang, X., Abaker Omer, A. A., Zhang, Z., et al. (2021). Polymer multilayer film with excellent UV-resistance and high transmittance and its application for glass-free photovoltaic modules. *Sol. Energy Mater. Sol. Cells* 229, 111103. doi: 10.1016/j.solmat.2021.111103

Li, M., Tao, K., Lu, J., Xu, S., Sun, Y., Chen, Y., et al. (2025). Preparation and properties of flexible multilayered transparent conductive films on substrate with high surface roughness. *Materials* 18, 3389. doi: 10.3390/ma18143389

Musselman, K. P., Uzoma, C. F., and Miller, M. S. (2016). Nanomanufacturing: High-Throughput, cost-effective deposition of atomic Scale thin films *via* atmospheric pressure spatial atomic layer deposition. *Chem. Mater.* 28, 8443–8452. doi: 10.1021/acs.chemmater.6b03077

Nagai, H., Hori, M., Goto, T., Fujii, T., and Hiramatsu, M. (2003). Fabrication of multilayered SiOCH films with low dielectric constant employing layer-by-layer process of plasma enhanced chemical vapor deposition and oxidation. *Jpn. J. Appl. Phys.* 42, 2775–2779. doi: 10.1143/JJAP.42.2775

Nallusamy, S., Vasanthi, V., Kavinkumar, T., Bhaviripudi, V. R., Mangalaraja, R. V., Srinivasan, R., et al. (2025). Advances in optoelectronics for environmental and energy sustainability. *Next Energy* 9, 100387. doi: 10.1016/j.nxener.2025.100387

Nguyen, V. H., Papanastasiou, D. T., Resende, J., Bardet, L., Sannicolo, T., Jiménez, C., et al. (2022). Advances in flexible metallic transparent electrodes. *Small* 18, 2106006. doi: 10.1002/smll.202106006

Nur-E-Alam, M., Vasiliev, M., and Alameh, K. (2020). "Dielectric/metal/dielectric (DMD) multilayers: growth and stability of ultra-thin metal layers for transparent heat regulation (THR)," in *Energy Saving Coating Materials* (Elsevier), 83–112. doi: 10.1016/B978-0-12-822103-7.00004-2

Orlenson, V. B., Akhramovich, L. N., and Volvach, A. E. (2022). Broadband absorption enhancement of metal-dielectric structures in the optical range and their application in solar energy storage. *Appl. Phys. Lett.* 120, 084105. doi: 10.1063/5.0075103

Parida, B., Baloch, A. A. B., Albadwawi, O., Adothu, B., Aly, S. P., Alberts, V., et al. (2026). Solution and vacuum-based scalable deposition methods for Perovskite/Si tandem solar cells. *Adv. Opt. Mater.* 14, e03341. doi: 10.1002/adom.202503341

Pastuszak, J., and Węgierek, P. (2022). Photovoltaic cell generations and Current research directions for their development. *Materials* 15, 5542. doi: 10.3390/ma15165542

Pontes, F. M., Leite, E. R., Lee, E. J. H., Longo, E., and Varela, J. A. (2001). Dielectric properties and microstructure of SrTiO3/BaTiO3 multilayer thin films prepared by a chemical route. *Thin Solid Films* 385, 260–265. doi: 10.1016/S0040-6090(01)00772-6

Sayem, A. S.Md., Lalbakhsh, A., Esselle, K. P., Buckley, J. L., O'Flynn, B., and Simorangkir, R. B. V. B. (2022). Flexible transparent antennas: advancements, challenges, and prospects. *IEEE Open J. Antennas Propag.* 3, 1109–1133. doi: 10.1109/OJAP.2022.3206909

Sengupta, A., Afroz, M. A., Sarkar, J., and Satapathi, S. (2026). Engineering of a near-infrared transparent dielectric–metal–dielectric electrode for *p – i – n* perovskite solar cells in 4T Silicon/Perovskite tandem photovoltaics. *ACS Appl. Mater. Interfaces* 18, 6701–6711. doi: 10.1021/acsami.5c18157

Sharfstein, S. T. (2022). Bio-hybrid electronic and photonic devices. *Exp. Biol. Med.* 247, 2128–2141. doi: 10.1177/15353702221144087

Sharma, V., Singh, S., Asokan, K., and Sachdev, K. (2016a). A study on 100 MeV O7+ irradiated SnO2/Ag/SnO2 multilayer as transparent electrode for flat panel display application. *Nucl. Instrum. Methods Phys. Res. B* 379, 141–145. doi: 10.1016/j.nimb.2016.04.059

Sharma, V., Vyas, R., Bazylewski, P., Chang, G. S., Asokan, K., and Sachdev, K. (2016b). Probing the highly transparent and conducting SnO_x/Au/SnO_x structure for futuristic TCO applications. *RSC Adv.* 6, 29135–29141. doi: 10.1039/C5RA24422F

Sharma, V., Kumar, P., Kumar, A., Asokan, K., and Sachdev, K. (2017). High-performance radiation stable ZnO/Ag/ZnO multilayer transparent conductive electrode. *Sol. Energy Mater. Sol. Cells* 169, 122–131. doi: 10.1016/j.solmat.2017.05.009

Sharma, V., Sharma, H., Singh, S. K., Kumar, R., Kumari, Y., and Sachdev, K. (2021). Organic–Inorganic hybrid structure as a conductive and transparent layer for energy and optoelectronic applications. *ACS Appl. Electron. Mater.* 3, 1601–1609. doi: 10.1021/acsaelm.0c01099

Sharma, S., Prabhu, S. S., and Gupta, B. K. (2026). From atomic layers to moire superlattices: engineering quantum interfaces in 2D heterostructures for next generation terahertz optoelectronics. *Small Methods* 10, e02426. doi: 10.1002/smtd.202502426

Sharmile, N., Chowdhury, R. R., and Desai, S. (2025). A comprehensive review of quality control and reliability research in Micro–Nano technology. *Technol. (Basel)* 13, 94. doi: 10.3390/technologies13030094

Shen, S., Wang, J., Zhu, Y., Yang, W., Gao, R., Li, J.-F., et al. (2022). Large-area metal-dielectric heterostructures for surface-enhanced raman scattering. *Opt. Express* 30, 38256. doi: 10.1364/OE.464631

Singh, S., Sharma, V., Saini, D., Asokan, K., and Sachdev, K. (2017). Fabrication of highly efficient TiO2/Ag/TiO2 multilayer transparent conducting electrode with N ion implantation for optoelectronic applications. *Ceram. Int.* 43, 9759–9768. doi: 10.1016/j.ceramint.2017.04.152

Singh, S., Sharma, V., Saini, D., Shekhawat, S., Asokan, K., and Sachdev, K. (2018). Influence of 100 keV Ar+ implantation on electrical and optical properties of TiO2/Ag/TiO2 multilayer films. *Mater. Sci. Semicond. process.* 75, 18–25. doi: 10.1016/j.mssp.2017.11.016

Skafi, Z., and Brown, T. M. (2025). "Conventional substrates for flexible devices," in *The Handbook of Paper-based Sensors and Devices* (Cham: Springer Nature Switzerland), 27–48. doi: 10.1007/978-3-031-91080-7_2

Sun, G., Shahid, M., Fei, Z., Xu, S., Eisner, F. D., Anthopolous, T. D., et al. (2019). Highly-efficient semi-transparent organic solar cells utilising non-fullerene acceptors with optimised multilayer MoO_3/Ag/MoO_3 electrodes. *Mater. Chem. Front.* 3, 450–455. doi: 10.1039/C8QM00610E

Tang, J. X., Lee, C. S., and Lee, S. T. (2006). Chemical bonding and electronic structures at magnesium/copper phthalocyanine interfaces. *Appl. Surf. Sci.* 252, 3948–3952. doi: 10.1016/j.apsusc.2005.09.027

Tauc, J. (1974). "Optical properties of amorphous semiconductors," in *Amorphous and Liquid Semiconductors* (Boston, MA: Springer US), 159–220. doi: 10.1007/978-1-4615-8705-7_4

Thabit, H. A., Kabir, N. A., and Ahmed, N. M. (2021). Synthesis and thermoluminescence characteristics and structural and optical studies of ZnO/Ag/ZnO system for dosimetric applications. *J. Lumin.* 236, 118097. doi: 10.1016/j.jlumin.2021.118097

Thallapally, P. K., Peter McGrail, B., Dalgarno, S. J., Schaef, H. T., Tian, J., and Atwood, J. L. (2008). Gas-induced transformation and expansion of a non-porous organic solid. *Nat. Mater.* 7, 146–150. doi: 10.1038/nmat2097

Triolo, C., Lorusso, A., Masi, S., Mariano, F., Della Torre, A., Accorsi, G., et al. (2025). Electromagnetic mode management in transparent DMD electrodes for high angular color stability in white OLEDs. *ACS Photonics* 12, 2413–2422. doi: 10.1021/acsphotonics.4c01956

Uyanik, Z., Turkoglu, F., Koseoglu, H., Ekmekcioglu, M., Ata, B., Demirhan, Y., et al. (2022). Enhanced optoelectronic properties of magnetron sputtered ITO/Ag/ITO multilayers by electro-annealing. *J. Vac. Sci. and Technol. B* 40, 042204. doi: 10.1116/6.0001868

Wang, T., Lu, K., Xu, Z., Lin, Z., Ning, H., Qiu, T., et al. (2021). Recent developments in flexible transparent electrode. *Cryst. (Basel)* 11, 511. doi: 10.3390/cryst11050511

Wang, Y., Nian, Y., Wang, S., Lu, C., Yin, L., Wang, C., et al. (2025). Recent progress in Dielectric/Ag/Dielectric transparent electrodes on flexible substrates. *Coatings* 15, 1370. doi: 10.3390/coatings15121370

Wu, J.-Y., An, B.-L., Dong, W., Yang, Z., and Duan, Y.-Y. (2024). Design, preparation, and property analysis of metal/dielectric multilayer film with wavelength selectivity. *J. Phys. Condens. Matter* 36, 255703. doi: 10.1088/1361-648X/ad33f0

Xiao, H., Long, C., Tian, X., and Chen, H. (2016). Effect of thorium addition on the thermophysical properties of uranium dioxide: atomistic simulations. *Mater. Des.* 96, 335–340. doi: 10.1016/j.matdes.2016.02.019

Yakimets, I., MacKerron, D., Giesen, P., Kilmartin, K. J., Goorhuis, M., Meinders, E., et al. (2010). Polymer substrates for flexible electronics: achievements and challenges. *Adv. Mat. Res.* 93–94, 5–8. doi: 10.4028/www.scientific.net/AMR.93-94.5

Yao, Z.-F., Cordova, D. L. M., Milligan, G. M., Lopez, D., Allison, S. J., Kuang, Y., et al. (2024a). Lattice-guided assembly of optoelectronically active π-conjugated peptides on 1D van der Waals single crystals. *Sci. Adv.* 10, eadl2402. doi: 10.1126/sciadv.adl2402

Yao, Z., Kuang, Y., Wu, H., Lundqvist, E., Fu, X., Celt, N., et al. (2024b). Selective induction of molecular assembly to tissue-level anisotropy on peptide-based optoelectronic cardiac biointerfaces. *Adv. Mater.* 36, 2312231. doi: 10.1002/adma.202312231

Yao, Z.-F., Lim, S., Kuang, Y., Lundqvist, E. M., Celt, N., Chung, C. O., et al. (2025). Complementary biomolecular coassemblies direct energy transport for cardiac photostimulators. *Proc. Natl. Acad. Sci.* 122, e2509467122. doi: 10.1073/pnas.2509467122

Yuan, Z., He, G., Li, S. X., Misra, R. P., Strano, M. S., and Blankschtein, D. (2022). Gas separations using nanoporous atomically thin membranes: recent theoretical, simulation, and experimental advances. *Adv. Mater.* 34, 2201472. doi: 10.1002/adma.202201472

Zeng, X., Jiang, X., Ning, Y., Gao, Y., and Che, R. (2024). Constructing Built-In electric fields with semiconductor junctions and schottky junctions based on Mo–MXene/Mo–Metal sulfides for electromagnetic response. *Nanomicro Lett.* 16, 213. doi: 10.1007/s40820-024-01449-7

Zhang, D., Yu, W., Zhang, L., and Hao, X. (2023). Progress in the synthesis and application of transparent conducting film of AZO (ZnO:Al). *Materials* 16, 5537. doi: 10.3390/ma16165537

Zhao, G., Shen, W., Jeong, E., Lee, S.-G., Yu, S. M., Bae, T.-S., et al. (2018). Ultrathin silver film electrodes with ultralow optical and electrical losses for flexible organic photovoltaics. *ACS Appl. Mater. Interfaces* 10, 27510–27520. doi: 10.1021/acsami.8b08578

Zhao, Y., Zhang, S., Fan, Y., Liu, Y., Zhang, Y., Jiang, L., et al. (2026). Multiscale manufacturing of organic electronic materials. *Adv. Mater.*, e09949. doi: 10.1002/adma.202509949

Zhou, K., Qi, B., Liu, Z., Wang, X., Sun, Y., and Zhang, L. (2024). Advanced organic–inorganic hybrid materials for optoelectronic applications. *Adv. Funct. Mater.* 34, 2411671. doi: 10.1002/adfm.202411671

Zhu, M.-Q., Jin, H.-D., Bi, P.-Q., Zong, F.-J., Ma, J., and Hao, X.-T. (2016). Performance improvement of TiO_2/Ag/TiO_2 multilayer transparent conducting electrode films for application on photodetectors. *J. Phys. D. Appl. Phys.* 49, 115108. doi: 10.1088/0022-3727/49/11/115108